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Deep Generative Graphs for Synthesizing Microstructure of Topology-Preserved Polymer-Bonded Explosives

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Abstract We introduce a generative artificial intelligence (AI) algorithm trained to generate realistic three-dimensional microstructures of polymer-bonded explosives (PBX). We idealized the PBX as a two-phase composite, trained to generate grains within a polymeric binder matrix. We introduce a two-stage hierarchical algorithm in which a deep generative graph model is used to handle the topology of the grains in the PBX, while a latent diffusion model generates the 3D geometry of the individual grains. This approach enables us to consider each grain as a simply-connected body embedded in three-dimensional space, hence avoiding the otherwise necessary but challenging task of generating manifolds of complex topology. To generate PBX microstructures with realistic grain topology, we sample sub-domains of segmented micro-computed tomography (CT) images of a mock PBX with Idoxuridine (IDOX) grains to constitute a data set. Each sub-domain is represented by a node-weighted graph with the position, size, and shape descriptors stored as node features. The topological data set is then used to train a graph recurrent neural network algorithm (GraphRNN) that learns to sequentially add nodes and edges to form a weighted graph that describes the IDOX granular network, consisting of nodes with features connected by edges. With the global topology prescribed, the individual grains are generated by a conditional denoising diffusion probabilistic model that takes the node features as inputs and creates individual grains that fit the prescribed node features generated from the GraphRNN. Numerical examples are provided to analyze the similarity between the actual and synthetic microstructures. The synthetic PBX microstructures are provided for third-party validation.

Keywords PBX, GraphRNN, generative AI, denoising diffusion, microstructures

1 Introduction

The dynamic thermomechanical behaviors of polymer-bonded explosives (PBX) are significantly influenced by the morphology of individual grains and their microstructural topology. Over the past decades, a growing body of work has established many critical multiscale relationships between key microstructural attributes. For instance, it is now established that the grain size distribution and binder-to-grain ratio may affect the propagation of shock waves and lead to different ignition thresholds and detonation velocities (Skidmore et al., 2000). As such, these relationships between microstructural properties and performance are often used as guiding principles for manufacturing processes such as die pressing (Graff Thompson et al., 2005), mixing (Claydon, 2021) and additive manufacturing (Groven and Mezger, 2017; Yeager et al.,

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2020). However, since the microstructure of PBX can only be indirectly manipulated through the manufacturing process, determining the key microstructural attributes that lead to the high performance of the PBX is crucial. However, due to the cost and labor required to conduct shock experiments in the PBX with energetic grains, it is not always feasible to use experiments alone to acquire the necessary data to construct the response surface between performance and microstructural controlling parameters. As such, mesoscale grain-scale simulations on PBX could play an essential role in generating auxiliary data that addresses the data demand (Kohler et al., 2017; Kim et al., 2018; Duarte et al., 2020; Ma et al., 2021; Vlassis et al., 2022; Roy et al., 2022; Ma et al., 2023) or at least provide further insight to formulate hypotheses on how microstructural attributes (e.g., grains-size distribution and fabric) affect the shock response. To ensure the fidelity of these mesoscale studies, it is important to ensure that the material models are sufficiently capturing the essence of constitutive responses, and that the microstructures used to perform the sensitivity analyses or parametric studies are sufficiently representative of the real microstructure.

1.1 Literature on microstructure reconstructions

Conventionally, microstructures generated for multiscale analysis can be generated by random field algorithms (Koutsourelakis and Deodatis, 2006; Feng et al., 2014). Such algorithms are designed to make the reconstructed media match the binary-valued marginal probability distribution function and the two-point correlation function of the reference media. However, this approach is often only applied in two-dimensional image generation and does not necessarily create realistic-looking microstructures. Thovert et al. (2001), on the other hand, introduces an algorithm that generates polydisperse spheres such that the reconstructed numerical specimen percolates even at low porosity and introduces statistical measures of the skeleton of the pore space represented by graphs.

In recent years, generative artificial intelligence (AI), such as generative adversarial networks (GANs) (Yang et al., 2018; Chun et al., 2020), variational autoencoders (VAEs) (Sundar and Sundararaghavan, 2020), flow models (Mirzaee and Kamrava, 2025), and diffusion models (Sohl-Dickstein et al., 2015; Dhariwal and Nichol, 2021), have all shown great promise in inferring microstructure from training samples. In theory, many of these generative AI algorithms can generate synthetic PBX microstructures. For instance, Nguyen et al. (2022) has utilized deep reinforcement learning to fine-tune the hyperparameters of GANs to generate synthetic sandstone micro-CT images. Guan et al. (2021) on the other hand, design a GAN model using only a 2D slice image as input and compare the statistical measures of the flow network. However, in these cases, the generated images may contain regions with over-blurred areas that lack fine-grained details and realistic textures, despite the use of an elaborate hyperparameter fine-tuning algorithm (Nguyen et al., 2022). Vlassis and Sun (2023), on the other hand, uses a latent diffusion model to generate 2D microstructures with targeted properties. Vlassis et al. (2024) extends this latent diffusion model to three dimensions to generate realistic sand grains. Then, a physical simulator is used to pour the generated grains into a container, forming assemblies. However, creating an assembly with a physical simulator requires high-fidelity simulations for specific problems. For materials that are manufactured through a more complex process involving damage, fracture, plastic deformation, or phase transition, using simulations to create assemblies is not always a viable option. It is presumably possible to directly train latent diffusion models to generate realistic-looking microstructures. However, unlike most computer vision tasks that lead only to realistic-looking images, the microstructures we generated are intended for high-fidelity physics-based simulation tasks, such as extending material databases, quantifying uncertainty and sensitivities, and other simulation tasks that require realistic mechanical properties. In the case of polymer-bonded explosives (PBXs), where hot spots often interact within the grains, preserving the topological features of the granular microstructures may be critical to generate microstructures with realistic material properties. In other words, a more direct control over how grains should be distributed in the polymer matrix is needed. Furthermore, while latent diffusion models have achieved great success in generating complex geometries in simply connected topological spaces, creating materials with complex topologies, in particular, host matrices with a large number of inclusions, is not trivial (Hu et al., 2024).

1.2 Generative AI for Topology-preserving synthetic polymer-bonded composites

Given the need to preserve topological features while generating microstructures with realistically-shaped grains (either single crystal or polycrystalline grains, which is a length scale of material representation we do not attain in this work), we introduce a staggered generative AI algorithm that breaks down the generation tasks of PBX into two sub-tasks handled by two coupled generative algorithms, i.e., (1) a MicrostructureRNN, our domain-specific adaptation of the GraphRNN autoregressive model, that generates the topology of synthetic polymer-bonded composites, where features and locations of the grains are prescribed, and (2) a latent diffusion model that generates individual grains with conditions introduced to fit the features prescribed by the GraphRNN, as illustrated in Figure 1.

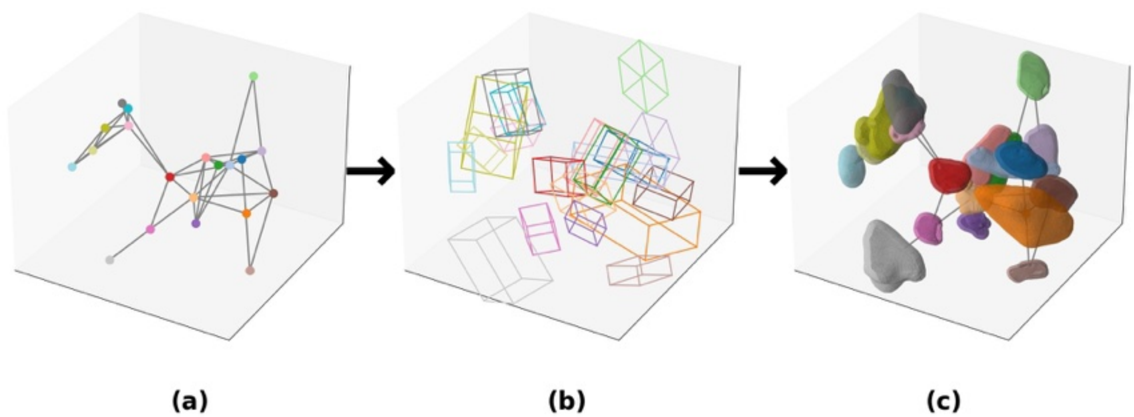


Fig. 1: Schematic of our hierarchical generation process to achieve synthetic 3D microstructures of PBX: (a) A MicrostructureRNN constructs the topology as a node-weighted graph, (b) where the node weights including coordinates, oriented bounding box (OBB), size, and a roughness measure are used to represent key features of the grain geometry, (c) a conditional diffusion model to generate consistent, high-fidelity grain geometries consistent with the assigned key features.

This division of tasks ensures that the structural realism of the PBX granular microstructure is preserved at both the global (connectivity) and local (grain geometry) levels. It serves as an ideal input for conditioning diffusion-based geometric synthesis. Given these advantages, we employ GraphRNN as the core topological generator in our framework. The following section outlines the architecture and implementation of GraphRNN for generating PBX microstructures.

1.3 Organization of the paper

The rest of this article is organized as follows. We will first describe the workflow of the multiscale generative AI algorithm, as illustrated in Figure 2. In particular, we examine the data curation step that generates the descriptors, i.e., the oriented bounding box (OBB), of the grains used to enforce the interaction between the topological generative AI and the geometry generative AI (Section 2. We then introduce the topological generation synthetic PBX weighted graphs enabled by the deep generative graph modified from (You et al., 2018) (Section 3). The descriptors prepared in Section 2 are then used as conditions for the conditional generation of individual grains via a latent diffusion model. The mechanism of this multi-scale coupling between the generative graphs and the latent diffusion will be described in Section 4. The numerical experiments conducted via mock IDOX polymer-bonded explosives and the conclusion of this research are provided in Sections 5 and 6 accordingly.

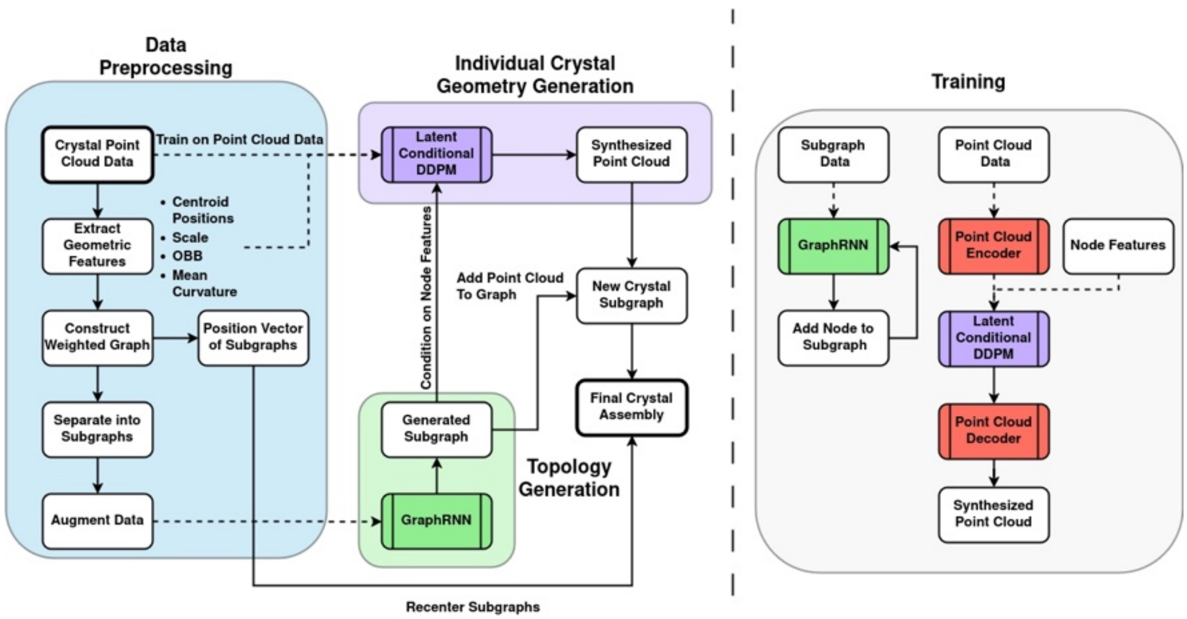


Fig. 2: Workflow for generating synthetic IDOX grain assemblies using the MicrostructureRNN (adapted from GraphRNN), a conditional DDPM, and a Point Cloud Autoencoder. The process begins by extracting geometric features from grain point cloud data (blue) to construct a weighted graph, which is then decomposed into subgraphs and augmented (see Section 2). GraphRNN (green) generates new subgraphs (see Section 3) while a Point Cloud Autoencoder (red) maps point cloud data into a lower-dimensional latent space. The Latent Conditional Diffusion Model (purple) operates within this learned latent space (see Section 4) to generate synthesized point clouds, which are subsequently decoded back into the original space. The pipeline iteratively integrates generated subgraphs and synthesized point clouds to assemble realistic granular microstructures.

2 Data Acquisition and Curation

This section introduces the data acquisition and curation procedures for preparing the necessary training data for specific representations in machine learning (ML) tasks. In all numerical examples, the micro-CT images of the mock IDOX assemblies are used as training data. This raw data is stored as voxelized images. For the MicrostructureRNN, we need to extract weighted graphs from the voxelized images to represent the topology of the IDOX assemblies (see Section 2.2). To train the DDPM, we convert the individual voxelized images into point clouds for the surface of each IDOX grain. This data curation step is closely aligned with the one used in Vlassis et al. (2024), except for the following modifications. For each voxelized image, we generate a dense point cloud for the surface, with surface normals estimated via local plane fitting using the `point cloud utils` package. We then enforced spatial uniformity through blue-noise downsampling and applied random Euler rotations for data augmentation. This yielded 11,892 training samples, each encoded as a $(1000, 6)$ tensor of 3D coordinates and surface normals. All point clouds were centered at the origin and normalized to ensure stable encoding of grain shapes in the autoencoder's latent space. To enable the topology-geometry coupling between the MicrostructureRNN and the DDPM, the

GraphRNN generates synthetic weighted graphs with descriptors of grains as node/edge weights, while the DDPM takes these descriptors as additional inputs for the score estimator of the DDPM to constrain the shape, size, and orientation of the synthetic grains. The descriptors that enabled the interaction between the two generative AIs are described in Section 2.3.

2.1 Synchrotron micro-computed tomography (SMT) Imaging

Computed tomography (CT) emerged in the 1990s as a powerful non-destructive 3D imaging technique to characterize materials using industrial CT scanners. X-ray radiation used in industrial CT scanners is generated by high-voltage X-ray tubes, resulting in an X-ray beam with a wider energy spectrum (polychromatic beam). This beam has limited resolution and requires long scanning durations due to the limited flux rates. GSECARS bending magnet beamline 13D (13 BM-D) at the Advanced Photon Source (APS), Argonne National Laboratory, Illinois, USA, offers the ability to acquire fast SMT 3D images using a synchrotron light source (Meisenheimer et al., 2020). The synchrotron light sources used in SMT allow for a higher flux rate of photons, which leads to a better spatial resolution and a better signal-to-noise ratio (Elnur and Alshibli, 2023), which eliminates beam hardening artifacts (Thompson et al., 1984), and which allows for shorter scanning durations compared to most conventional industrial CT systems.

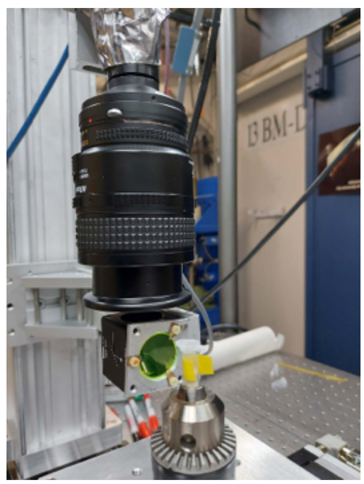


Fig. 3: Plastic vials placed on the stage of beamline 13D at APS .

The 3D SMT images reported in this paper were acquired using magnet beamline 13D (13 BM-D, APS. The IDOX crystals were placed inside plastic vials with an outer diameter of 8 mm and an inner diameter of 7 mm (See Figure 3). The configuration resulted in an X-ray beam with an energy ranging from 21 keV to 62 keV. To acquire the 3D images, specimens were scanned at exposure times ranging between 7 and 9 milliseconds. The stage configuration consists of an Aerotech hexapod, an Aerotech air-bearing rotation stage, and a Huber X-Y translation stage. The rotation stage was set to rotate from 0^o to 180^o, triggering the camera (FLIR Grasshopper3 with a maximum frame rate of 163 frames/s) at 0.1^o increments. A Nikon Macro lens with 0.223 and 0.225 focus was connected to the camera. A 0.25-mm-thick non-coated pure lutetium aluminum garnet (LuAG) scintillator was used to convert the X-ray radiation into visible light quickly and efficiently. The acquired scans were initially processed using tomoDisplay software for dark field correction, flat field correction, zinger removal, and image (volume) reconstruction. The SMT images have a spatial image resolution = 3.35 μm/pixel. The raw 3D SMT grayscale images were processed using preprogrammed modules as part of Thermo Fisher Scientific's PerGeos commercial software. The images were first filtered using an anisotropic diffusion filter module to enhance the contrast and reduce image noise. The grayscale pixel value range for IDOX grains was identified based on the data histogram, and the images were binarized by assigning a value of 1 to IDOX grains and a value of 0 to air voids using an interactive thresholding module.

2.2 Node set partitions from a Micro-CT image

To train the GraphRNN, we generate a dataset with 25,000 subgraphs from a single IDOX granular assembly. The resulting dataset was split 80/20 for training and validation. The raw IDOX dataset is provided in the form of 10,480 STL files, one for each grain derived from high-resolution micro-CT scans. We remove the STL files that do not exhibit sufficient resolution or contain non-manifold geometries, and retain 3,964 unique samples that accurately reflect the morphological complexity of the original assembly. Each file also includes the position and orientation information, which allows us to reconstruct the IDOX assemblies.

We then partitioned the sets of grains using repeated K-Means clustering on node positions, varying both the number of clusters k and random seeds, i.e.,

$$\min_{S} \sum_{i=1}^{k} \sum_{\mathbf{x} \in S_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|^2 \tag{1}$$

where k is the number of clusters;; $S = \{S_1, S_2, \ldots, S_k\}$ denotes the set of clusters, each containing data points; $\mathbf{x}$ represents a data point; $\boldsymbol{\mu}_i$ denotes the centroid (mean) of the data points belonging to cluster S_i, and $\|\mathbf{x} - \boldsymbol{\mu}_i\|^2$ is the squared Euclidean distance between the data point and its cluster centroid. The k-mean clustering objective function Eq. (1) is minimized iteratively by assigning points to the nearest centroid, while updating centroids based on current cluster assignments, i.e.,

$$S_i^{(t)} = \{\mathbf{x} : \|\mathbf{x} - \boldsymbol{\mu}_i^{(t)}\|^2 \le \|\mathbf{x} - \boldsymbol{\mu}_j^{(t)}\|^2, \forall j, 1 \le j \le k\} \tag{2}$$

$$\boldsymbol{\mu}_i^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{\mathbf{x} \in S_i^{(t)}} \mathbf{x} \tag{3}$$

These steps are repeated until convergence, at which time assignments no longer change. In our case, we ran the K-means clustering with K=50, 100, and 200. The resultant node sets for the training data *mathbbV* are shown in Fig. 4.

2.3 Construction of node/edge weighted subgraphs

Once the set of grains in the micro-CT images is partitioned into node sets $\mathbb{V}$, additional information is supplemented to further describe the assemblies. In particular, we introduce an edge set $\mathbb{E}$ that describes the spacing, the set of node weights that describe the geometrical attributes of individual grains, and a set of edge weights, $\mathbf{E}_w$ that stores the Euclidean distance of grain pairs. Together, these four sets form a 4-tuple or a node/edge weighted undirected graph $G = (\mathbb{V}, \mathbb{E}, \mathbf{X}, \mathbf{E}_w)$ that represents the topology of the sampled sub-domains of the IDOX assembly.

2.3.1 Node Weights: descriptors of grain geometry

In this work, eleven node weights are handpicked to represent the geometrical attributes of the grains. These features include the centroid position, size and aspect ratio, orientation (represented by the oriented bounding box)and averaged mean curvature. These features, summarized in Table 1, are normalized and then concatenated as a feature vector. For each node $v_i \in \mathbb{V}$, there exists a corresponding 11-dimensional feature vector, which is stored as the i-th row of the node feature matrix $\mathbf{X}$.

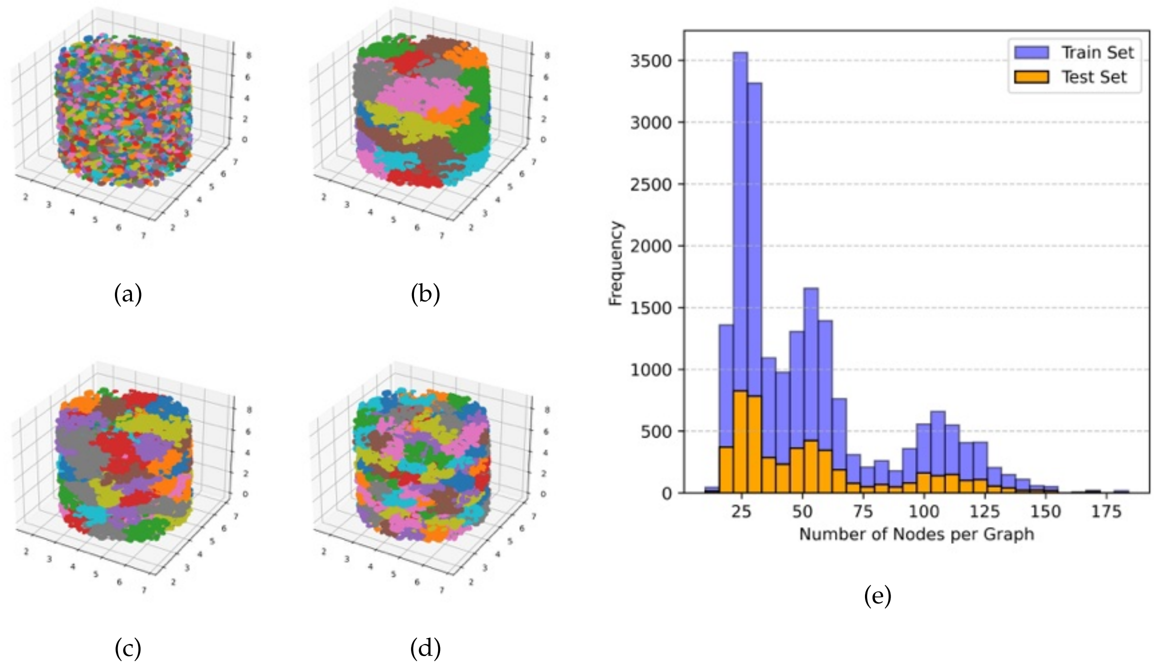


Fig. 4: Subgraph sampling. (a) Original IDOX assembly. (b–d) Subgraphs from K-means clustering with $k =$ 50, 100, and 200. (e) Node count distribution reflects structural diversity from varied k-means clustering scales.

Table 1: Summary of node weights in a microstructure graph

Feature	**Dim.**
Centroid Position	3
Effective Radius	1
Oriented Bounding Box (OBB)	6
Roughness measure	1

Note that each node in the graph G represents a grain in the host matrix. For convenience, the geometry of the grain is represented by a point cloud on the surface, where the spacing between points on the surface is approximately equal. This equal spacing, which is achieved through Poisson sampling disk (cf. Wei (2008); Williams et al. (2019)), enables us to obtain the centroid position by averaging the positions of individual points in the Cartesian coordinates. The **Centroid** defines the spatial location of a grain and is represented by its 3D coordinates $(x, y, z) \in \mathbb{R}^3$. We denote this spatial position of node i as $p_i \in \mathbb{R}^3$. Anchoring each grain in space enables the model to capture position-dependent relationships within the microstructure. The **Effective Radius** $r \in \mathbb{R}^+$ is defined as the maximum distance from the centroid to a point at the grain surface (Barua et al., 2013).

A **Roughness Measure** is inferred from the reconstructed vertex normals, following Botsch et al. (2010). For a triangle with vertices $\mathbf{x}_i, \mathbf{x}_j, \mathbf{x}_k$, the triangle normal $\mathbf{n}_T$ is:

$$\mathbf{n}_T = \frac{(\mathbf{x}_j - \mathbf{x}_i) \times (\mathbf{x}_k - \mathbf{x}_i)}{\|(\mathbf{x}_j - \mathbf{x}_i) \times (\mathbf{x}_k - \mathbf{x}_i)\|}. \tag{4}$$

Vertex normals $\mathbf{n}_v$ are the weighted averages (angle-based) of adjacent face normals in the one-ring neighborhood $N_1(\mathbf{v})$ Botsch et al. (2010); Porteous (2001):

$$\mathbf{n}_v = \frac{\sum_{T \in N_1(\mathbf{v})} \alpha_T \mathbf{n}_T}{\| \sum_{T \in N_1(\mathbf{v})} \alpha_T \mathbf{n}_T \|}, \tag{5}$$

where α_T are angle-based weights Botsch et al. (2010). Normals are propagated from vertices and preserved on the spatially uniform point cloud obtained via Poisson-disk downsampling.

To measure surface roughness from the point cloud, we construct a local shape operator $\mathbf{S}(p_i)$ at each point p_i using its $K = 10$ nearest neighbors. A local tangent frame is derived by applying Singular Value Decomposition (SVD) to the centered neighborhood; the two right-singular vectors orthogonal to the normal define the tangent directions $\mathbf{e}_1, \mathbf{e}_2$.

Each neighbor offset $p_j - p_i$ and normal difference $\mathbf{n}_j - \mathbf{n}_i$ is projected into this frame. The normal derivatives $\partial \mathbf{n}/\partial u$ and $\partial \mathbf{n}/\partial v$ are approximated via least-squares fits (Rusinkiewicz, 2004):

$$\begin{bmatrix} u_j \ v_j \end{bmatrix} \mapsto -\begin{bmatrix} \Delta n_{u,j} \ \Delta n_{v,j} \end{bmatrix}, \tag{6}$$

yielding the 2×2 shape operator matrix $\mathbf{S}$, which is symmetrized to ensure real eigenvalues. This operator linearly maps in-plane directions to changes in normal direction, encoding local curvature. The neighborhood size K balances local sensitivity and robustness to noise.

The principal curvatures $k_1(p_i), k_2(p_i)$ are obtained as the eigenvalues of $\mathbf{S}(p_i)$, representing the maximum and minimum bending directions. Mean curvature at each point is defined as:

$$H(p_i) = \frac{k_1(p_i) + k_2(p_i)}{2}. \tag{7}$$

The global roughness measure is then computed as the average absolute mean curvature over the $N = 1000$ sampled points:

$$H_{\text{mean}} = \frac{1}{N} \sum_{i=1}^{N} |H(p_i)|. \tag{8}$$

Using $|H|$ ensures that both convex and concave regions contribute positively, avoiding cancellation between opposing curvatures.

Equations (7)–(8) define a compact differential geometry-based descriptor compatible with our generative framework, allowing controlled modulation of the complexity of the grain surface of PBX.

The **Oriented Bounding Box (OBB)** is a rectangular box, arbitrarily oriented, that tightly encloses a 3D point set. It was initially formulated by O'Rourke as a minimal-volume enclosure (O'Rourke, 1985), later popularized for efficient collision detection (Gottschalk, 2000), and approximated for computational efficiency via PCA (Chang et al., 2011). In this study, the OBB serves as both an auxiliary descriptor in the graph G's node weights and as a conditioning parameter for the latent diffusion model.

An OBB is characterized by two components: (1) the spatial extents $(d_x, d_y, d_z) \in \mathbb{R}^+ \times \mathbb{R}^+ \times \mathbb{R}^+$, normalized as $d_x^2 + d_y^2 + d_z^2 = 1$, ensuring grain shape is represented distinctly from scale (with effective radius and positions included separately), and (2) its orientation (ϕ, θ, ψ) specified by intrinsic ZYX (yaw–pitch–roll) Euler angles.

We compute the box by performing PCA on the grain's point cloud $\mathbf{P} = \{p_i \in \mathbb{R}^3\}$ using Open3D (Zhou et al., 2018). Denoting the principal axes (columns of $\mathbf{R} \in SO(3)$) as u_x, u_y, u_z, the full widths are

$$d_k = \max_{p \in P}(u_k \cdot p) \; - \; \min_{p \in P}(u_k \cdot p) \; = \; \max_{p_i, p_j \in P} \left| u_k \cdot (p_i - p_j) \right|, \quad k \in \{x, y, z\}, \tag{9}$$

or,

$$(d_x, d_y, d_z) = \max_{p_i, p_j \in P} \left| \mathbf{R}^\top (p_i - p_j) \right|. \tag{10}$$

This PCA-based heuristic matches the fast approximation strategy in (Chang et al., 2011) and underlies Open3D's `get_oriented_bounding_box()` (Open3D Development Team, 2025). We extract the Tait–Bryan Euler angle (ϕ, θ, ψ) from $\mathbf{R}$ via

$$\phi = \arctan \frac{R_{32}}{R_{33}}, \quad \theta = \arctan \frac{-R_{31}}{\sqrt{R_{32}^2 + R_{33}^2}}, \quad \psi = \arctan \frac{R_{21}}{R_{11}}, \tag{11}$$

using the ZYX convention (Diebel, 2006; SciPy Community, 2025). This Tait–Bryan Euler angle is compatible with the standard intrinsic rotation conventions and provides consistent mapping between rotation matrices and grain orientations. To prevent gimbal lock, we constrain angles using SciPy's `as_euler('zyx')` convention and constrain the angles as (Slabaugh, 2000) (see Fig. 5):

$$X\ (\phi) : [-180^\circ, +180^\circ], \quad Y\ (\theta) : [-90^\circ, +90^\circ], \quad Z\ (\psi) : [-180^\circ, +180^\circ].$$

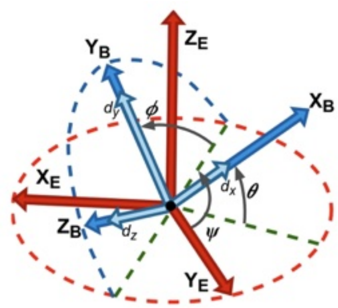


Fig. 5: Tait–Bryan angles (ZYX convention) illustrating sequential rotations: yaw (ψ) about Z_E, pitch (θ) about Y_B, and roll (ϕ) about X_B. The initial Earth-fixed frame (red) is rotated into the Body-fixed frame (blue). The dimensions (d_x, d_y, d_z), shown in light blue, represent the axis-aligned extents of the OBB.

2.3.2 *Edge and edge weights: Spacing between grain-pairs*

For granular assemblies, grain contacts can be represented by an edge set that defines the physical interaction between two contacting grains (Satake, 1992; Wang and Sun, 2017; Vlassis et al., 2024). For composites with granular inclusions, the inclusions may still interact with each other, with or without the physical contacts (Dandekar and Koslowski, 2021; Ravindran et al., 2023). As such, we generate an edge set based on a hypothesis that there are cut-off distances beyond which the interaction between two inclusions in the host matrix is negligible compared to those of other inclusion pairs. As such, we introduce an ad hoc rule that edges are assigned to at least $k_{min} = 5$ neighbors for each vertex in the graph that represents the inclusion. The edge is then obtained from the classical nearest neighborhood algorithm described in Alg. 1. Finally, each edge can be assigned the Euclidean distance between the centroids of the grain pair as the edge weight.

Algorithm 1 Scale-Adaptive Radius Graph Construction. Grain pairs (i, j) are connected by an edge if their distance is within a scale-based adaptive radius, ensuring physically realistic connectivity.

Require: grain positions p_i and scales s_i for all grains $i \in \{1, \ldots, n\}$
Require: Scale factor $f \leftarrow 3$
Ensure: Graph $G = (\mathbb{V}, \mathbb{E}, \mathbf{X}, \mathbf{E}_w)$ with edges determined by adaptive radius criterion
Initialize node set $\mathbb{V} \leftarrow \{1, \ldots, n\}$ and empty edge set $\mathbb{E} \leftarrow \emptyset$
for all grain pairs (i, j) with $i < j$ **do**
    Compute distance: $d_{ij} \leftarrow \|p_i - p_j\|$
    Compute adaptive radius: $R_{ij} \leftarrow f \cdot (s_i + s_j)$
    if $d_{ij} \leq R_{ij}$ **then**
        Add edge (i, j) to $\mathbb{E}$ with weight $w_{ij} \leftarrow d_{ij}$
    end if
end for
return $G = (\mathbb{V}, \mathbb{E}, \mathbf{X}, \mathbf{E}_w)$

3 MicrostructureRNN: Topological Generation of PBX Microstructures

We introduce *MicrostructureRNN*, a domain-specific adaptation of GraphRNN (cf. You et al. (2018) and Popova et al. (2019)) tailored to synthesize PBX microstructures that preserve topological features relevant for polymer-bonded explosives. In particular, we revise the original model by incorporating heteroscedastic negative log-likelihood (NLL) losses to train an autoregressive recurrent neural network that handles the heteroscedastic noises of different features.

3.1 MicrostructureRNN Architecture for Generating Microstructures

Given a set consisting of M observed PBX graphs $\mathbb{G} = \{G_1, G_2, ...G_M$, the model learns the underlying distribution $p(\mathcal{G})$ that enables the generation of new samples $G_{\text{new}} \sim p(\mathcal{G})$ by introducing a recurrent neural network to generate nodes and edges sequentially consistent with the data set $\mathbb{G}$. Following autoregressive sequence models (Graves, 2013), we use recurrent networks to parameterize each conditional factor and train by maximizing the log-likelihood of observed nodes and edges.

The probability of the existence of a graph G is the product of the node and edge conditional probabilities. Through the chain rule, the joint probability distribution reads,

$$p(G) = \prod_{i=1}^{N} p(x_i \mid x_{1:i-1}) \prod_{j=1}^{i-1} p(w_{ij} \mid w_{1:i-1,\cdot}, X_{\leq i}), \tag{12}$$

where $w_{ij} \in [0,1]$ are normalized continuous edge weights (not binary adjacency). As such, we transform the graph generation problem into a sequence generation problem. Due to the conditional probability terms in Eq. (12), the order of the appearance of nodes and edges must be specified such that the node ordering is consistent.

3.1.1 Breadth-First Search for autoregressive model

We adopt a Breadth-First Search (BFS) traversal from GraphRNN (Das et al., 2023; Zhu et al., 2022) to impose a consistent node ordering π while reducing extra steps otherwise necessary for edge generation (cf. You et al. (2018)). The BFS takes a random permutation as input. Then it picks a node as the tree root, then expands the tree structure by assigning the neighboring nodes of the root in the next level, before expanding the tree further to the next level. This step is repeated until there are no more nodes left. The resultant tree leads to a sequence S^{π} that reduces the number of steps required for edge generation by a proper node ordering π. The connectivity of a graph G with n nodes is first represented by an adjacency matrix $A_{\pi} \in \{0,1\}^{n\times n}$, where $A_{\pi_{i,j}} = 1$ iff $(\pi(v_i), \pi(v_j)) \in E$. $W_{\pi} \in [0,1]^{n\times n}$ is the continuous edge-weight matrix (normalized distances). The adjacency sequence is $S_i^{\pi} \in [0,1]^{i-1}$, where each entry stores the normalized weight $w_{\pi(k),\pi(i)}$.

Recall that the autoregressive model employs sequential learning to facilitate training. As such, a recurrent neural network composed of gated recurrent units (GRUs) (Chung et al., 2014) is used. However, other sequential learning techniques, such as a 1D convolutional neural network and a transformer, if trained successfully, can also be used for a function. The order of the appearance of each node and edge is represented by a sequence of *edge-weight vectors* $S^{\pi} = (S_1^{\pi}, \dots, S_n^{\pi})$ (see Fig. 6):

$$S_i^{\pi} = (w_{1,i}^{\pi}, \dots, w_{i-1,i}^{\pi})^T, \tag{13}$$

where each entry $w_{k,i}^{\pi} \in [0,1]$ is the normalized continuous edge weight from node $\pi(k)$ to node $\pi(i)$ (set to zero if no edge exists after adaptive-radius filtering). This sequence of edge-weight vectors is used as the label data for supervised learning.

3.1.2 Recurrent neural networks for node and edge generations from Gaussian distributions

There are two forecasting RNN models built and run sequentially, i.e., (1) NodeRNN, which takes the previous subgraph and adds a new node, and (2) EdgeRNN predicts each entry of S_i^π as a Gaussian distribution, outputting mean $\mu_{ij} \in [0,1]$ (via Sigmoid) and variance σ_{ij}^2, modeling continuous edge weights. By encoding graphs as adjacency sequences, this reformulation recasts graph generation as a conditional sequence modeling task:

$$p(S^\pi) = \prod_{i=1}^{n+1} p\left(S_i^\pi \mid S_1^\pi, S_2^\pi, \ldots, S_{i-1}^\pi\right) = \prod_{i=1}^{n+1} p\left(S_i^\pi \mid S_{<i}^\pi\right), \tag{14}$$

where the special end-of-sequence token (EOS) is represented as node $n+1$. Both RNN models are retained via supervised learning such that the conditional probabilities in Equation (12) given each adjacency vector S_i^π can be predicted. These predictions are supervised using ground truth graphs and optimized with heteroscedastic Gaussian NLL losses on node features and normalized continuous edge weights, see Equations (15)–(16) Murphy (2012). For NodeRNN we assume each ground-truth feature vector $\boldsymbol{x}_i^{GT} \in \mathbb{R}^D$ is drawn from a Gaussian distribution with predicted mean and variance $(\mu_i, \sigma_i^2) = g_\theta(h_i^{\text{node}})$, where $h_i^{\text{node}} = \text{NodeRNN}(h_{i-1}^{\text{node}}, X_{<i})$.

$$\mathcal{L}_{\text{node}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{2}\frac{\|\boldsymbol{x}_i^{GT} - \mu_i\|^2}{\sigma_i^2} + \frac{1}{2}\log(2\pi\sigma_i^2)\right], \tag{15}$$

where $\|\cdot\|$ is the L2 norm for the normalized feature vector $\boldsymbol{x}$ and N is the number of nodes in the graph. This loss function corresponds to the negative log-likelihood of a Gaussian with heteroscedastic (sample-dependent) variance (cf. Graves (2013)).

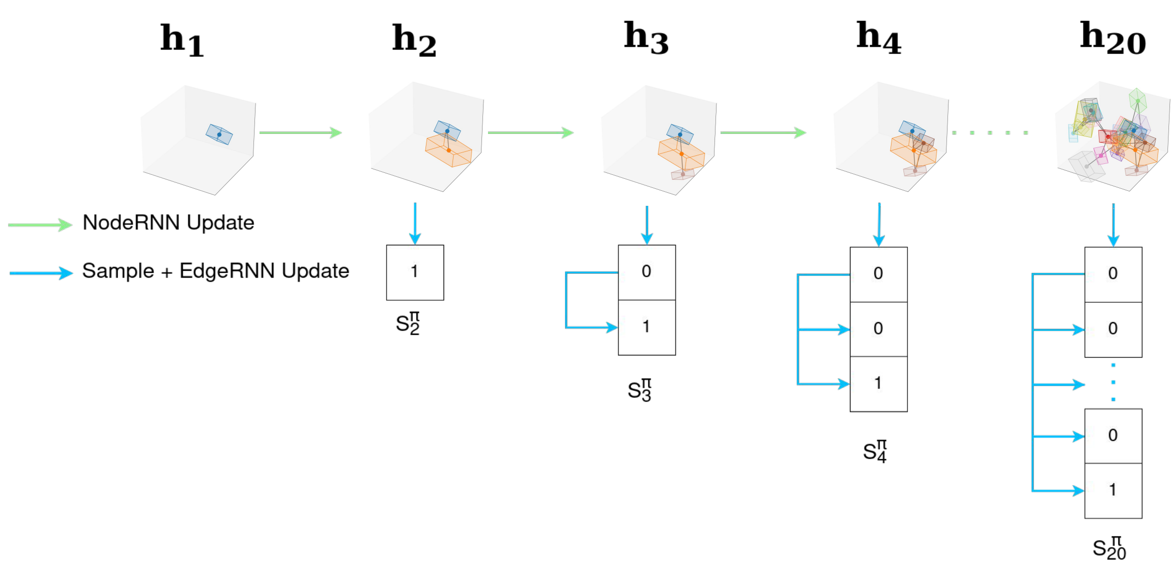


Fig. 6: GraphRNN generation process: the NodeRNN sequentially adds nodes representing grains, including generative priors such as oriented bounding boxes (OBBs), while the EdgeRNN predicts their connectivity based on local graph topology and spatial proximity. This stepwise procedure updates the continuous edge-weight vectors S_i^π (later thresholded to binary adjacency), each predicting new edges conditioned on the current partial graph state (see Equation (12)).

For periodic attributes such as Euler angles, features are represented as $(\sin, \cos)$ pairs; Gaussian NLL is applied in $\mathbb{R}^2$ and renormalized at sampling. Positive attributes such as radii are predicted in log-space. OBB extents are predicted in unconstrained logits and normalized to satisfy geometric constraints. For EdgeRNN, we assume each ground-truth edge weight A_{ij}^{GT} is again drawn from a Gaussian distribution with predicted mean $\hat{A}_{ij}$ and variance σ_{ij}^2, both output by the EdgeRNN:

$$\mathcal{L}_{\text{edge}} = \frac{1}{\sum_{i=1}^{M}(i-1)} \sum_{i=1}^{M} \sum_{j<i} \left[\frac{1}{2} \frac{(w_{ij}^{GT} - \mu_{ij})^2}{\sigma_{ij}^2} + \frac{1}{2} \log(2\pi\sigma_{ij}^2) \right], \tag{16}$$

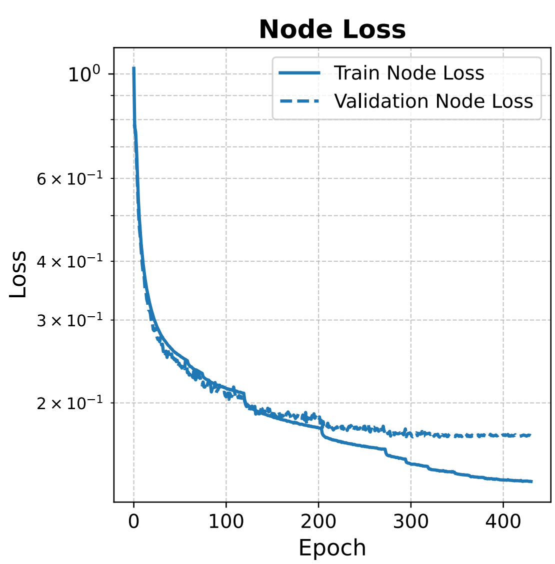
where M is the number of nodes and the loss is applied only to valid edges (masking out pairs outside the adaptive radius). We model normalized edge weights in $[0, 1]$ using a Gaussian likelihood, a standard and effective choice for regression tasks in machine learning (Bishop, 2006). While alternatives such as the Beta (Johnson et al., 1995; Ferrari and Cribari-Neto, 2004) or logit-Normal (Atchison and Shen, 1980) distributions explicitly respect bounded support, we found the Gaussian formulation to be stable, efficient, and sufficiently accurate in practice. These two neural networks are co-trained as shown in Eq. (12). The corresponding multi-objective loss is a weighted sum of node and edge terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{node}} \mathcal{L}_{\text{node}} + \mathcal{L}_{\text{edge}} + \lambda_{\text{var}} \sum_{i=1}^{N} (\log \sigma_{\text{node},i}^2 + \log \sigma_{\text{edge},i}^2), \tag{17}$$

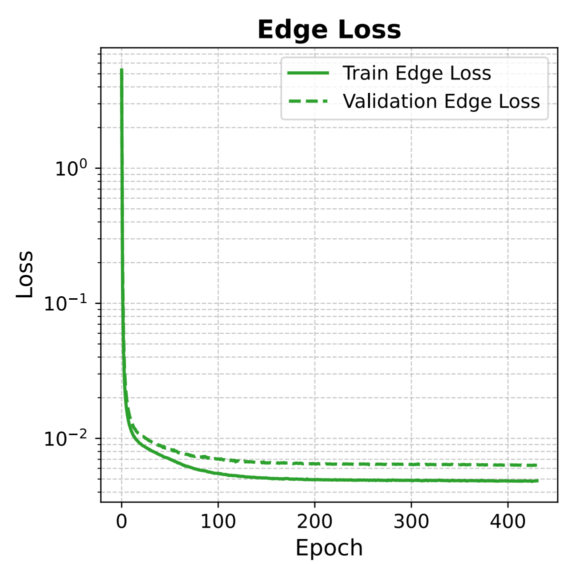
where $\lambda_{\text{node}} = 5.0$, $\lambda_{\text{var}} = 0.001$, and all $\log \sigma^2$ terms are clamped to $[-5, 5]$. Here N denotes the total number of training samples used in each mini-batch.

3.2 Architecture of MicrostructureRNN

MicrostructureRNN employs a sequential recurrent design with two modules: a **NodeRNN** that generates nodes and an **EdgeRNN** that assigns their connections, preserving graph topology.



(a) NodeRNN loss.



(b) EdgeRNN loss.

Fig. 7: Training and validation loss curves showing stable convergence of GraphRNN.

The NodeRNN embeds each feature vector $x \in \mathbb{R}^F$ via a two-layer MLP (LayerNorm, ReLU), concatenates the embedding with x, and propagates it through a 4-layer GRU (256 hidden units, dropout 0.15) and multihead self-attention. Outputs are $f = 5$ Gaussian heads $(\mu_j, \log \sigma_j^2)$ predicting 14 node attributes, with $\log \sigma^2 \in [-5, 5]$ and variance regularization $\lambda_{\text{var}} = 0.001$. The EdgeRNN projects the NodeRNN hidden state, embeds scalar edge features (two-layer MLP), and advances through a 4-layer GRU (64 hidden units, dropout 0.15). Edge weights follow

$$\hat{e} = \sigma(W_2 \tanh(W_1 h)), \tag{18}$$

with loss restricted to valid lower-triangular pairs. Hidden states evolve autoregressively,

$$h_i^{\text{node}} = \text{NodeRNN}(h_{i-1}^{\text{node}}, X_{<i}), \qquad e_{k,i} \sim \text{EdgeRNN}(h_{i,0}^{\text{edge}}; e_{k,i}),\ k < i. \tag{19}$$

We use AdamW (Loshchilov et al., 2017), a variant of the Adam optimizer that improves weight decay regularization, to train both the NodeRNN and EdgeRNN, with learning rates 3×10^{-4} (NodeRNN) and 1.5×10^{-5} (EdgeRNN), weight decay 10^{-3}, and a ReduceLROnPlateau scheduler (patience=10, factor=0.5). The loss from training over 400 epochs (see Figure 7) indicates stable convergence, with node-loss decreasing from approximately 1.2 to 0.14 on training and 0.17 on validation sets, and edge-loss decreasing from 4.0 to roughly 10^{-2} (train) and 10^{-3} (validation). These metrics demonstrate that MicrostructureRNN successfully learns realistic connectivity patterns and spatial relationships within PBX microstructures.

4 Conditional Denoising diffusion of grains represented by point clouds

To improve computational efficiency, we conduct the denoising diffusion process in the latent space generated by a point cloud autoencoder. This setting enables us to directly conduct the denoising process in the ambient space, which can be memory- and computationally intensive. The resultant latent-space conditional diffusion framework consists of an autoencoder that embeds the grain geometry represented by point clouds onto a low-dimensional space, and a conditional diffusion model that generates new geometries guided by topological features generated by the MicrostructureRNN for a specific graph vertex. Figure 8 diagrams the end-to-end pipeline: encoding, conditioned diffusion, and decoding of 3D grain shapes.

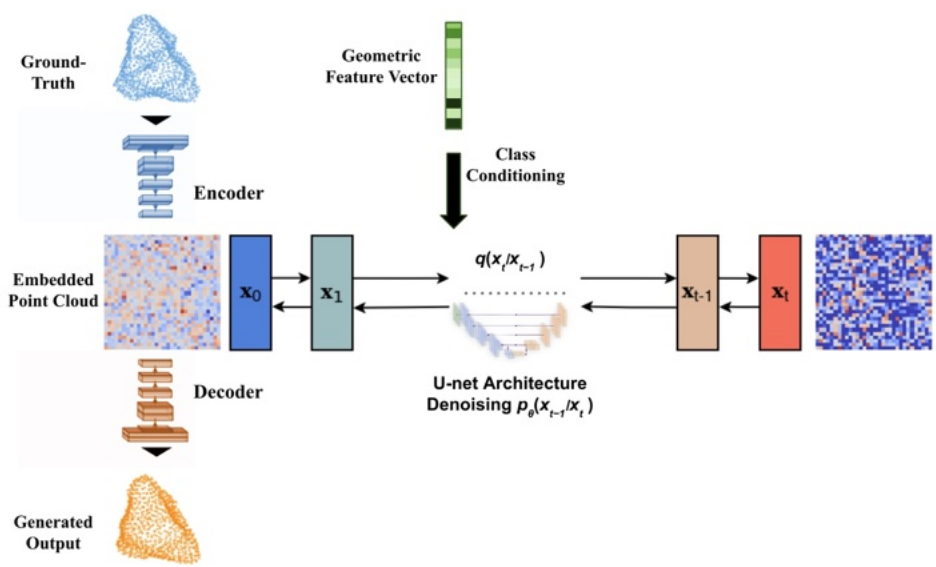


Fig. 8: **Overview of the IDOX grain generation process.** The point cloud autoencoder encodes real IDOX grains into a latent space, where the conditional DDPM iteratively refines their structure through stochastic denoising. GraphRNN-derived topological features are passed as conditions to the diffusion model to guide this stochastic process.

4.1 Point Cloud Autoencoder

The autoencoder architecture extends standard point cloud autoencoders by integrating geometry-aware parallel encoding, feature fusion, lightweight self-attention, and residual decoding layers, building on principles from PointNet and recent geometric deep learning methods (Qi et al., 2017; Zhao et al., 2019; Vaswani et al., 2017). We adapt the feature fusion and attention mechanisms from standard geometric learning practices, drawing inspiration from PointWeb and Transformer architectures (Zhao et al., 2019; Vaswani et al., 2017; He et al., 2016). The encoder extends the PointNet architecture to process each point cloud $P = \{p_1, \ldots, p_N\} \subset \mathbb{R}^6$, where each point p_i includes 3D coordinates and surface normals (Qi et al., 2017). Surface normals support geometry-aware regularization. The resulting latent code is a 32×32 feature map (from a 1024D vector), preserving spatial locality and enabling the diffusion model to synthesize shapes with greater topological coherence than flat embeddings (Vlassis et al., 2024).

The autoencoder is trained against the Chamfer Distance $\mathcal{L}_{\text{chamfer}}$ and an additional constraint $\mathcal{L}_{\text{normal}}$ designed to generate consistent surface normal between the original and reconstructed point clouds (Achlioptas et al., 2018), i.e.,

$$\mathcal{L}_{\text{total}} = \lambda_p \cdot \mathcal{L}_{\text{chamfer}} + \mathcal{L}_{\text{normal}} \tag{20}$$

where chamfer loss is scaled by a factor of $\lambda_p = 10$ to balance its influence relative to the normal loss, i.e.,

$$\mathcal{L}_{\text{chamfer}} = \frac{1}{|\mathcal{P}|} \sum_{p_i \in \mathcal{P}} \min_{q_j \in \mathcal{Q}} \|p_i - q_j\|_2^2 + \frac{1}{|\mathcal{Q}|} \sum_{q_j \in \mathcal{Q}} \min_{p_i \in \mathcal{P}} \|q_j - p_i\|_2^2 \tag{21}$$

$$\mathcal{L}_{\text{normal}} = \frac{1}{|\mathcal{P}|} \sum_{p_i \in \mathcal{P}} \left(1 - |\cos\theta(p_i, q_{\text{nn}(i)})|\right) + \frac{1}{|\mathcal{Q}|} \sum_{q_j \in \mathcal{Q}} \left(1 - |\cos\theta(q_j, p_{\text{nn}(j)})|\right) \tag{22}$$

where $\mathcal{P}$ and $\mathcal{Q}$ are sets of output and target points, p_i and q_j are individual points, and $q_{\text{nn}(i)}$ is the nearest neighbor of p_i in $\mathcal{Q}$. The term $\|p - q\|_2^2$ denotes squared Euclidean distance, and $\cos\theta(p, q)$ is the cosine similarity between normals at those points. The denoising diffusion process is then conducted in the latent space generated by the point-cloud autoencoder to cut the computational cost, following Vlassis et al. (2024) and Xiao et al. (2025).

4.2 Conditional Latent Diffusion

We use the latent diffusion model to infer 3D grains from the latent space (Nichol and Dhariwal, 2021; Rombach et al., 2022). We define a linear noise schedule $\beta_t \in [0.0001, 0.025]$, and compute $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, following the variance-preserving formulation of Ho et al. (2020). We then use $T = 1000$ time steps with a linear $\beta_t \in [0.0001, 0.025]$ to train the model to conduct diffusion in reverse time. At the inference stage, classifier-free guidance follows the exponential schedule in Eq. 32, with $w_{\min} = 0.2$ and $w_{\max} = 0.7$.

Let $\mathbf{x}_0 \in \mathbb{R}^{32 \times 32}$ denote a latent representation of a PBX grain, encoded from point cloud data. The forward diffusion process progressively adds Gaussian noise over T steps. At each step, the transition is:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t}\mathbf{x}_{t-1}, \beta_t \mathbf{I}), \tag{23}$$

With variance schedule $\beta_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^{t}(1 - \beta_s)$, the resulting marginal distribution is:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}). \tag{24}$$

The reverse process is learned by a neural network $\epsilon_\theta(\mathbf{x}_t, t)$ that predicts added noise at each timestep, reparameterizing the mean as:

$$\boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) \right). \tag{25}$$

The denoising model ϵ_θ is implemented as a U-Net that maps a noisy latent embedding $\mathbf{x}_t \in \mathbb{R}^{32\times32}$ and timestep t (and optionally a conditioning vector y) to a predicted noise tensor of the same shape. We train the model using the standard noise prediction loss:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2\right], \tag{26}$$

sampling t uniformly and corrupting $\mathbf{x}_0$ with known noise and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ as Gaussian noise. During training, the model learns to denoise embeddings corrupted at different stages.

This denoising process approximates the reverse-time dynamics of a stochastic differential equation (SDE) (Song et al., 2020) governed by the score function $\nabla_x \log p_t(x)$:

$$dx = \left[f(x,t) - g(t)^2\nabla_x \log p_t(x)\right] dt + g(t)d\bar{w}. \tag{27}$$

where $\nabla_x \log p_t(x)$ is the time-dependent *score function*. The model ϵ_θ approximates this score (up to scale), and the loss $\mathcal{L}_{\text{simple}}$ corresponds to a discretized denoising score-matching objective. This aligns our approach with the variance-preserving (VP) SDE used in modern diffusion models. In practice, we follow the discrete formulation from DDPM (Ho et al., 2020), where the denoised latent at step $t-1$ is computed as:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z}, \tag{28}$$

where $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$ and σ_t controls the residual sampling noise. This step corresponds to an Euler discretization of the reverse SDE.

4.2.1 Classifier-Free Guidance

Our generative framework enables detailed control over grain geometry by conditioning the denoising diffusion process on geometric priors derived from the GraphRNN-generated topology. Generated node features, carrying OBBs, curvature, and effective radius, are passed as geometric priors to the diffusion model, guiding it to respect spatial alignment, size, and morphological constraints. This conditioning is crucial for preserving both global connectivity and local geometric realism in the synthesized microstructures. We use classifier-free guidance, which avoids the need for a separately trained classifier and simplifies training. While classifier guidance can enhance perceptual quality, it risks biasing results toward classifier-aligned features that may diverge from physical realism (Nichol et al., 2022). Classifier-free guidance combines conditional and unconditional scores from the generative model, making it a simpler and more reliable approach for physically grounded synthesis under explicit constraints (Ho and Salimans, 2022). Classifier-free guidance enables conditional generation without a separate classifier by training a single network for both conditional and unconditional denoising. During training, the conditioning vector y is randomly replaced by the empty set $\emptyset$, yielding

$$\epsilon_\theta(\mathbf{x}_t, t, y) = \epsilon_\theta(\mathbf{x}_t, t, y = \emptyset). \tag{29}$$

During the inference, conditioning is introduced through the score function,

$$\nabla_{\mathbf{x}_t} \log p(y|\mathbf{x}_t) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|y) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t), \tag{30}$$

which leads to the extrapolated denoising score and exponential decay schedule

$$\bar{\epsilon}_\theta(\mathbf{x}_t, t, y) = \epsilon_\theta(\mathbf{x}_t, t, y) + w(t)\big(\epsilon_\theta(\mathbf{x}_t, t, y) - \epsilon_\theta(\mathbf{x}_t, t)\big). \tag{31}$$

where

$$w(t) = w_{\min} + (w_{\max} - w_{\min})\exp(-\lambda t/T), \tag{32}$$

with $\lambda = 5$, $T = 1000$, and $w \in [0.2, 0.7]$, so that conditioning influence diminishes as denoising proceeds. This balances fidelity with adherence to priors, producing morphologically faithful PBX grains. Figure 9 illustrates how bounding-box and curvature constraints are enforced as constraints, y is used to generate grains that fit inside the OBB with consistent roughness.

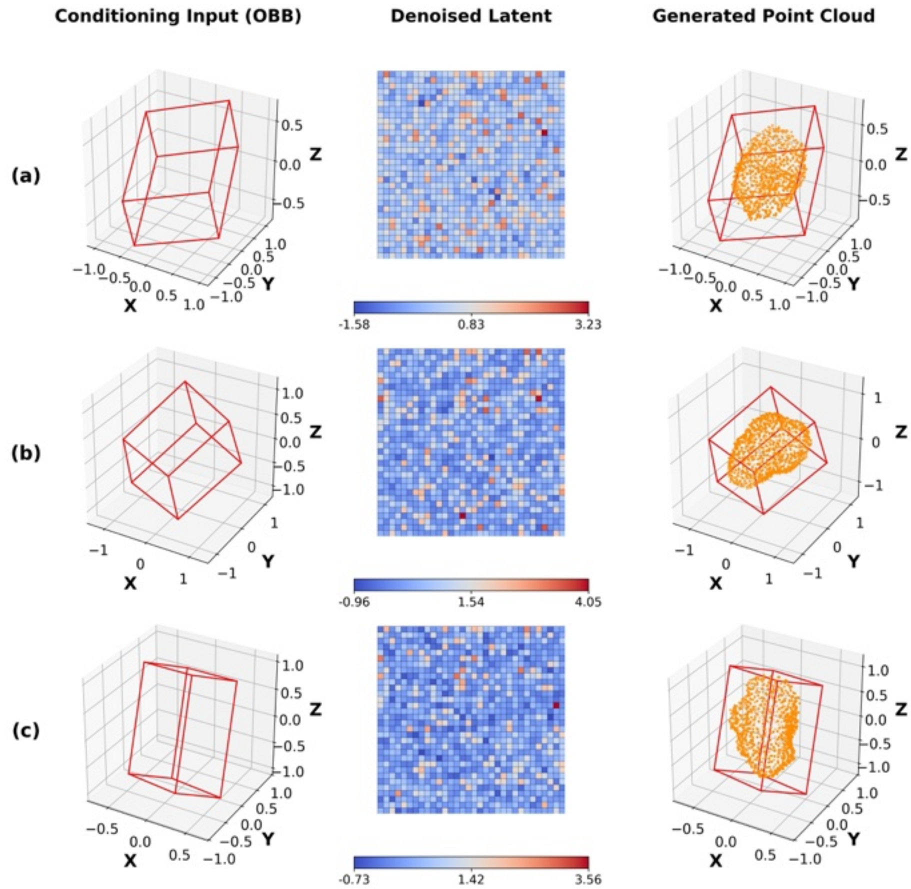


Fig. 9: **Bounding box constraints in classifier-free conditional DDPM generation.** node features containing specified oriented bounding box, effective radius, and mean curvature are used as class conditions for the conditional DDPM to control the shape of the generated point cloud to ensure natural geometric variations. The scale of the OBB is normalized for shape generations.

Remark 1 **Self-attention block.** A self-attention block with residual connections refines the fused representation. Features are pooled into a fixed latent grid for the diffusion model. After fusion, features are passed through a Conv1d-based self-attention block:

$$Q = W_Q X, \quad K = W_K X, \quad V = W_V X, \quad \mathrm{Attn}(X) = V \cdot \mathrm{Softmax}(Q^\top K) \tag{33}$$

where W_Q, W_K, W_V are 1×1 convolutions and the attention output is added to X (residual connection). In our implementation, this self-attention is single-head and uses `torch.bmm` to compute the attention weight matrix of size $[B, N, N]$, followed by a residual connection to preserve original features.

Remark 2 **Post-Processing Voxelization**. To evaluate mechanical response or fabricate microstructures, generated point clouds must be converted into watertight, voxelized models. We therefore apply Poisson surface reconstruction and voxelization using Open3D and custom Python tooling (Kazhdan et al., 2006; Zhou et al., 2018). Figure 10 illustrates the complete pipeline from point cloud to structured voxel model.

Each point cloud encodes coordinates and surface normals produced by the conditional diffusion model. We first rescale point clouds to physical units and apply Poisson surface reconstruction at depth 8,

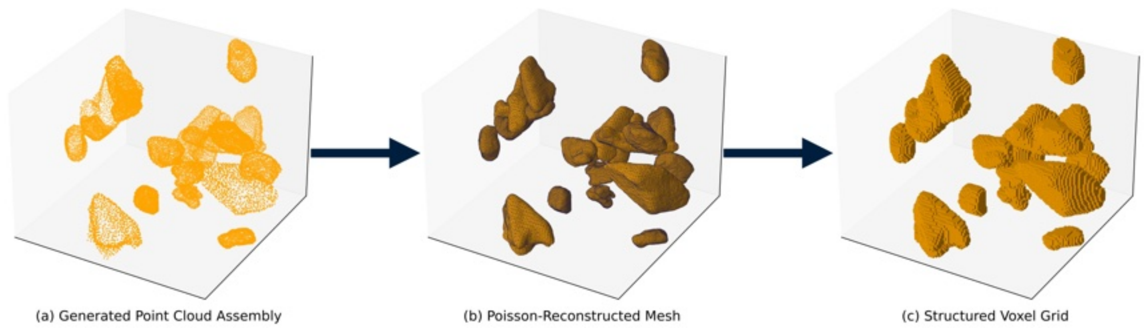


Fig. 10: From generated point clouds to structured voxel models for simulation.

balancing fidelity and computational cost. The resulting meshes are merged into a single watertight geometry and voxelized globally at 0.01-unit resolution to enforce spatial alignment and resolution uniformity. The voxelized models preserve geometric fidelity and export cleanly to CAD and simulation formats such as STL.

Remark 3 **Implementation details.** The autoencoder was implemented in PyTorch (Paszke, 2019) with 1D convolutions and adaptive pooling; chamfer loss was computed via PyTorch3D (Ravi et al., 2020); voxelization and OBB reconstruction used Open3D (Zhou et al., 2018); and preprocessing relied on NumPy (Harris et al., 2020).

5 Results: Evaluation of Synthetic Microstructures

In this section, we examine whether the two-stage generative model is capable of generating synthetic PBX microstructures with consistent statistics, both in terms of the topology of the granular constituent in the PBX microstructure and the geometry of individual grains. Geometric fidelity is tested through validations of empirical cumulative distribution functions (eCDF) of the curvature, orientation tensors, compactness, flakiness, and elongation, and bounding-box alignment (see Section 5.1). Meanwhile, Section 5.2 analyzes the consistency of topology generated from the GraphRNN. Statistical data gathered from these two sections provides us with a set of quantitative measures on whether the synthetic microstructures preserve the physical traits of the micro-CT images.

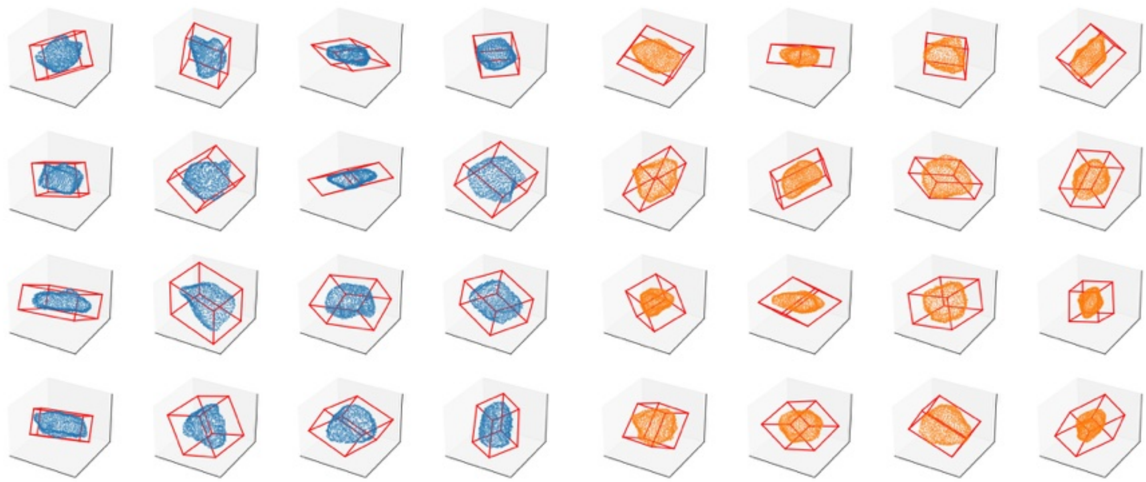
5.1 Grain Geometry and Morphology

This section examines whether our generative model reproduces these morphological traits consistent with the training data. Figure 11 provides a side-by-side comparison of the real and synthetic grains. In both cases, the oriented bounding box (OBB) constraints used to enforce conditional generation are also shown. This visual inspection reveals that all the grains are able to fit inside the OBB and that all six faces of the OBB are touched by each grain in the corresponding OBB.

5.1.1 Curvature and Bounding Box Controls

Grain morphology shapes mesoscale PBX behavior by governing local stress fields, fracture initiation, and packing geometry. Capturing both global grain structure and local surface variation is therefore essential for generating physically plausible surrogates. We quantify local surface variation by calculating the average of the mean curvature of all the points in the point cloud (see Eq. (7) and (8)).

To test the influence of the curvature conditions on the reverse diffusion process, we vary the curvature guidance signal within fixed OBBs. Figure 12 demonstrates that the DDPM generates rough and smooth



(a) Original Point Cloud Data (b) Generated Point Cloud Data

Fig. 11: Comparison of (a) micro-CT point cloud data with calculated oriented bounding boxes, and (b) generated point clouds with corresponding bounding boxes and curvature values. Visual similarity in curvature and bounding box fit suggests that the DDPM model replicates key morphological traits seen in micro-CT data.

variants without altering size or global geometry, confirming consistent roughness modulation across the latent space.

Figure 13(a) presents the eCDF of the curvature measure of the 5,000 generated grains and compared against the benchmark. Figures 13(b) and (c) compare the eCDF of OBB dimensions and Euler angles. In both metrics, real and generated samples align closely across most of the range. Mild compression at the tails indicates under-representation of extreme aspect ratios and rotational deviations—likely due to training priors or shape regularization. Together, these results confirm that the model preserves grain-scale morphological diversity while satisfying bounding box constraints. The statistical agreement across curvature and orientation metrics supports the geometric plausibility of the generated grains.

5.1.2 Quantitative Shape Analysis via Orientation Tensors

To compare the shapes of the real and generated grains, we first compute the surface orientation tensor (cf. Orosz et al. (2021)), then calculate their compactness, flakiness, and elongation. The second-order orientation tensor is defined as:

$$f_{ij} = \frac{1}{\sum_k A_k} \sum_k A_k n_i^k n_j^k, \tag{34}$$

where A_k is the area of face k and n^k its outward normal. The eigenvalues $f_1 \geq f_2 \geq f_3$ yield three dimensionless shape descriptors:

$$C = \frac{f_3}{f_1}, \quad F = \frac{f_1 - f_2}{f_1}, \quad E = \frac{f_2 - f_3}{f_1}, \tag{35}$$

representing compactness (C), flakiness (F), and elongation (E), respectively. These lie on a simplex: $C + F + E = 1$. Table 2 summarizes these metrics across 5,000 generated and 11,892 real grains. The generated distributions closely track the ground truth, with slightly higher compactness and elongation and reduced variance, consistent with mild regularization.

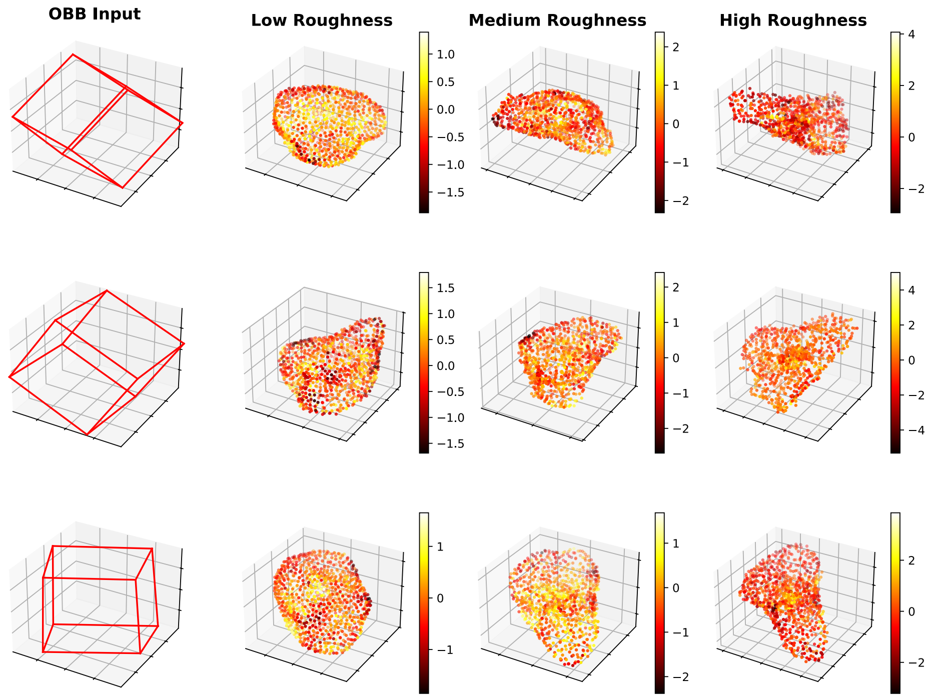


Fig. 12: **Conditional DDPM control of grain morphology via mean roughness.** Each row shows a fixed oriented bounding box (red, far left) and generated grains conditioned on increasing normalized mean curvature (0.00, 0.50, 1.00; left to right). Grains are colored using a `matplotlib` heatmap to highlight local curvature variations, illustrating precise morphological control essential for synthesizing realistic PBX microstructures.

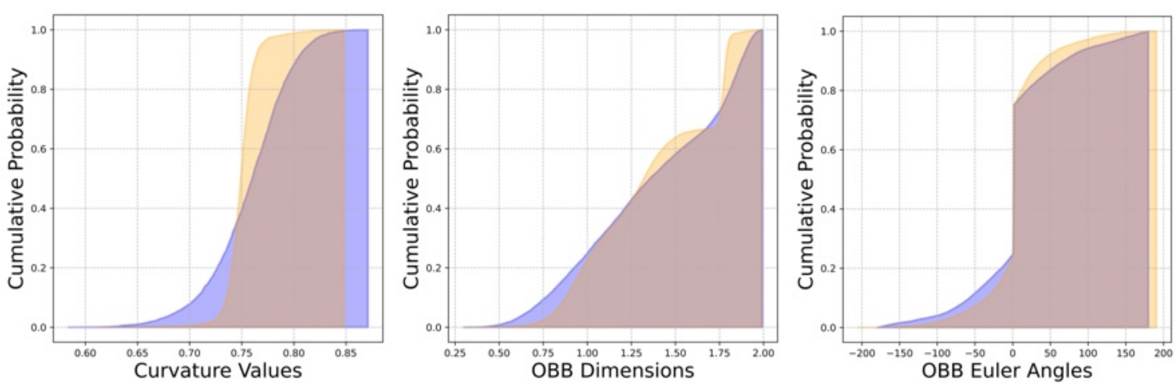


Fig. 13: The cumulative distribution functions (eCDFs) of surface mean curvature (left), oriented bounding box (OBB) dimensions (middle), and Euler angles (right) for real (blue) and generated (orange) grains. Close alignment across both distributions supports the model's fidelity in representing anisotropic shape and orientation, with only minor deviations at the tails.

Figure 14 plots these distributions in ternary form. The convex hulls of the real and generated grains overlap in most regions, except near the corner where $F = 1, E = 0$, and $C = 0$. Note that the conditional generation uses the oriented bounding box constraints, which should be sufficient to control the compactness, flakiness, and elongation. Nevertheless, this discrepancy still exists. We speculate that this slight shift toward the compact–elongate region could be attributed to the underrepresentation of highly flaky grains.

Table 2: Shape Metrics Comparison Between Real and Generated PBX Grains

Sample	$\mathbf{C_{mean}}$	$\mathbf{F_{mean}}$	$\mathbf{E_{mean}}$	$\mathbf{C_{median}}$	$\mathbf{F_{median}}$	$\mathbf{E_{median}}$	$\mathbf{C_{st.\ dev.}}$	$\mathbf{F_{st.\ dev.}}$	$\mathbf{E_{st.\ dev.}}$
Real Dataset	0.2568	0.5044	0.2470	0.2341	0.5080	0.2452	0.1269	0.2166	0.1048
Generated	0.2842	0.4441	0.2773	0.2667	0.4357	0.2743	0.1117	0.1782	0.0834

Due to the low dimensionality of the latent space, it is also possible that fine details of the grains can not be fully represented, and hence we observe a smoothening effect due to the data compression.

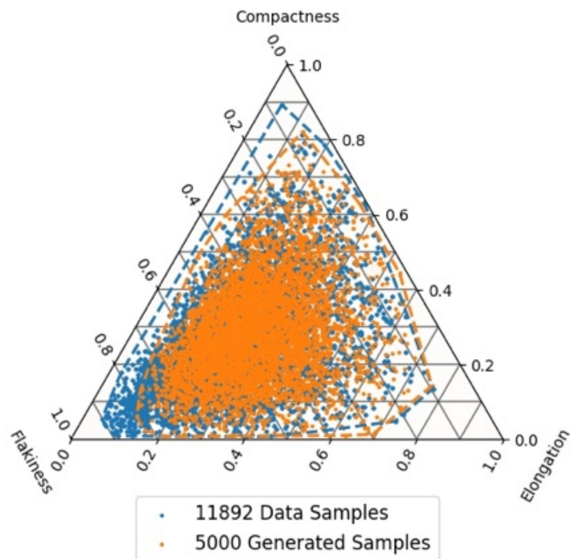


Fig. 14: Ternary plot of compactness (C), flakiness (F), and elongation (E) for real and generated grains. Convex hulls represent the distribution bounds for each set.

5.1.3 Grain Size Distribution

Grain size distribution (GSD) of a synthetic microstructure can be obtained from the set of OBBs inferred by the MicrostructureRNN. Generating the consistent grain size distribution is important for the PBX component, as it exhibits a strong influence on the mesoscale mechanisms that tie to the performance, i.e., effective wave propagation speed, void collapse, and ignition sensitivity (Field, 1992; Beckvermit et al., 2013). Accurate reproduction of GSD is therefore critical for generating physically realistic microstructures. We evaluate GSD using the effective radius

$$r_{\text{eff}} = \left(\frac{3V}{4\pi}\right)^{1/3}, \tag{36}$$

Figure 15 compares the empirical cumulative distribution function (eCDF) for real and generated grains. The synthetic samples show a narrower spread, clustering around mid-scale values. While this reflects generative stability, it under-represents the tails—suppressing both small and large grains and modestly reducing packing diversity. This central bias likely arises from latent priors and bounding box regularization, which constrain geometric extremes to ensure stable generation. While the effect is limited, excessive scale uniformity may suppress high-aspect-ratio pores and restrict void topology under shock. Nonetheless, the model accurately captures the bulk of the GSD, supporting its suitability for generating compositionally plausible microstructures for mesoscale simulation.

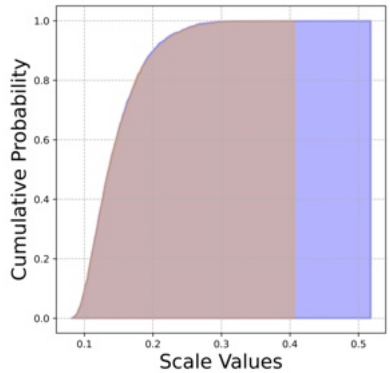
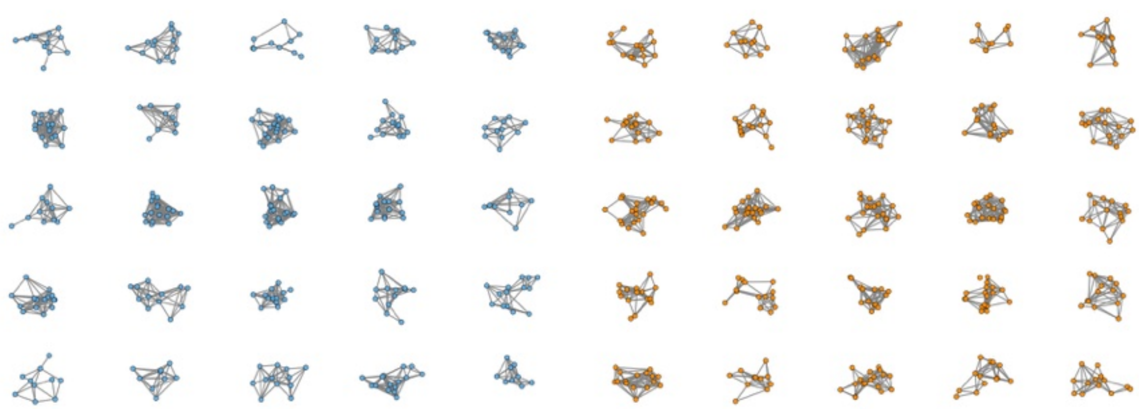


Fig. 15: The eCDF of grain scale values (mm) for original (blue) and generated (orange) PBX grains. The generated distribution is more peaked, favoring mid-scale grains.

5.2 Topological Structure and Spatial Connectivity

Accurate topology is critical to the mesoscale fidelity of PBX microstructures. While grain morphology governs local response, it is the topological arrangement that controls how stress, heat, and detonation energy propagate through the material (Van der Steen et al., 1989; Belon et al., 2020). This section evaluates whether the GraphRNN model generates physically plausible PBX topologies. We evaluate topology at three structural levels. Local topology is captured by edge-weight fidelity (Section 5.2.1), degree distributions (Section 5.2.2), and clustering coefficients (Section 5.2.3). Global connectivity is characterized by graph density (Section 5.2.4).



(a) Original Graph Data (b) Generated Graph Data

Fig. 16: Comparison of (a) real and (b) generated graph networks. Qualitative similarities in clustering, density, and local motifs suggest plausible structural reproduction.

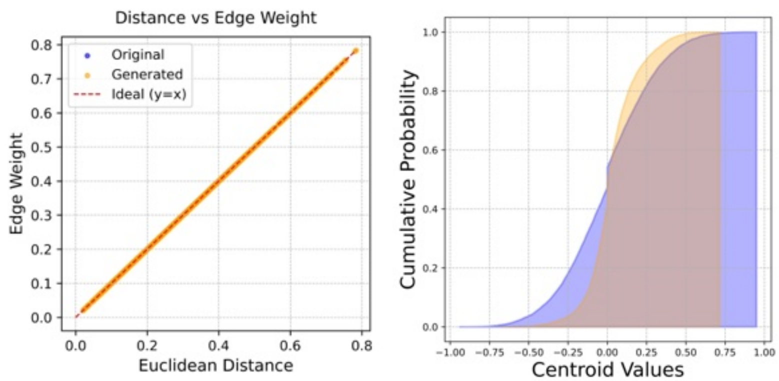
5.2.1 Inter-Grain Spacing (Edge Weights)

Inter-grain spacing governs mesoscale phenomena, such as hotspot formation, pore collapse, and energy localization in PBX microstructures (Field, 1992; Ravindran et al., 2023). Note that grain spacing can be quantified by the average of the edge weights, which are the Euclidean distance between the centroids of two grains, i.e.,

$$w(e_{ij}) = \|\mathbf{c}_i - \mathbf{c}_j\|_2,$$

where $\mathbf{c}_i$ is the centroid of grain i.

Figure 17 validates the model's spatial encoding. Panel (a) confirms that edge weights exhibit a strong 1:1 correlation with centroid distances, verifying geometric consistency. Panel (b) shows that centroid positions in the generated data align with the ground truth across the bulk domain, with slight compression at the tails likely due to latent space regularization.



(a) Edge weights vs. centroid distances (b) Centroid position distributions

Fig. 17: Validation of inter-grain spatial encoding. (a) Edge weights closely track centroid distances, confirming spatial fidelity. (b) Generated centroid positions match ground truth in the bulk domain, with mild compression at extremes.

We assess proximity statistics across entire graphs that represent the spacing of the grains. Recall that we assign the Euclidean distance as the edge weight. As such, we can analyze the spacing by calculating the mean and standard deviation.

$$\text{Mean Edge Weight} = \sum_{e_{ij} \in \mathbb{E}} \frac{w(e_{ij})}{N_e} \tag{37}$$

where N_e is the total number of edges. Figure 18 shows that early models (epoch 10) produce over-clustered graphs with low mean spacing. By epoch 250, the distribution converges to the real data, capturing both tight and loose configurations.

The standard deviation distribution (Figure 19) follows the same trend: early over-regularization is corrected by epoch 250, reproducing the full spread of local spacing variation.

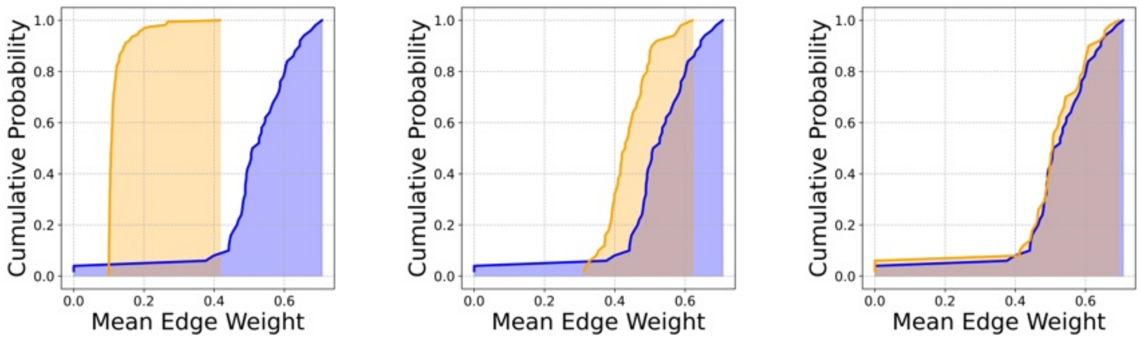
These results confirm that GraphRNN captures both the mean and variance of inter-grain spacing—critical for modeling heterogeneous features, such as dense agglomerates and loosely packed zones, that govern the detonation response.

5.2.2 Node Degree Distribution and Local Coordination

The degree of node, the number of neighboring grains with edges paired with the i-th grain, is defined as,

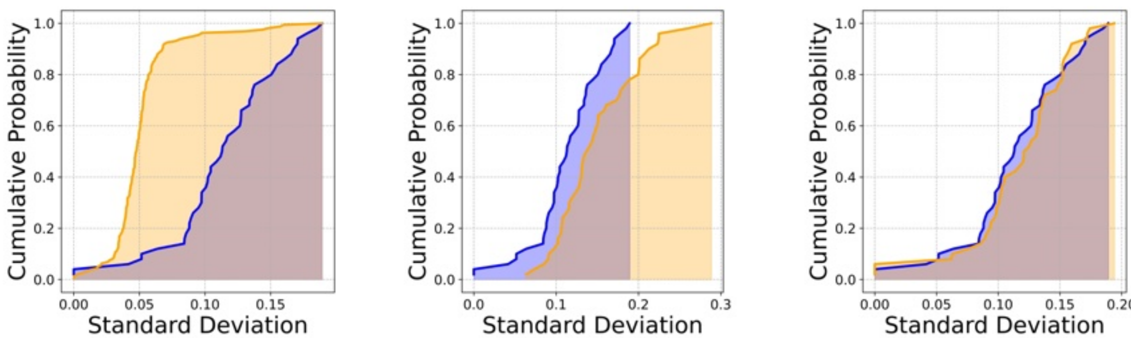
$$k_i = |\{j : e(i,j) \in \mathbb{E}\}|, \tag{38}$$

The degrees for every node in the graph can be computed by the row sum of the assembly matrix and the average of the diagonal terms of the degree matrix. Since we assign edges $e(i,j)$ on particular pairs within a threshold distance, the average degree of nodes is expected to be higher than the coordination



(a) Epoch 10 (b) Epoch 50 (c) Epoch 250

Fig. 18: The eCDF of mean edge weights across epochs. Graphs from later epochs increasingly align with the original PBX distribution, reflecting improved realistic spacing.



(a) Epoch 10 (b) Epoch 50 (c) Epoch 250

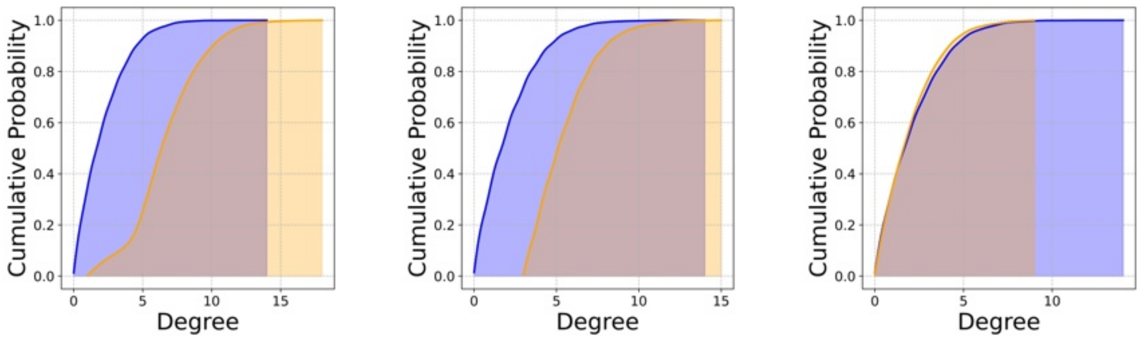
Fig. 19: The eCDF of the standard deviation of edge weights. The generated graphs converge toward the original data, capturing realistic heterogeneity in inter-grain spacing.

number of the contacted grains (Satake, 1992). Figure 20 tracks the cumulative distribution function of degree across training epochs. Graphs generated at the beginning of the training (epoch 10) exhibit more diverse degrees of connections and are generally more crowded (as indicated by the higher graph degrees than the benchmark). Toward the end of the training (epoch 250), the generated graphs converge toward the real distribution, with the mean being significantly closer than the benchmark. The slopes of the eCDFs also match well, indicating that the spacing of the generated and benchmark microstructures is consistent. Nevertheless, the eCDF of the generated microstructure reaches 100% at a lower degree than the benchmark. This indicates that the GraphRNN performs worse in generating the highly crowded regions of the microstructures.

Degree distribution offers a necessary but insufficient validation of topology. It captures local connectivity but not motif structure or energy pathways (Ravindran et al., 2023; Kim et al., 2018). Subsequent metrics (clustering, spectral density) assess these higher-order structural traits.

5.2.3 Triadic Clustering and Local Cohesion

Local cohesion in PBX structures depends on clustering, measuring the probability that a grain's neighbors are themselves connected, capturing local motifs critical to energy retention and stress buildup in PBX



(a) Epoch 10 (b) Epoch 50 (c) Epoch 250

Fig. 20: Progression of cumulative distributions of degree over training. GraphRNN corrects early over-coordination, matching real local connectivity by epoch 250.

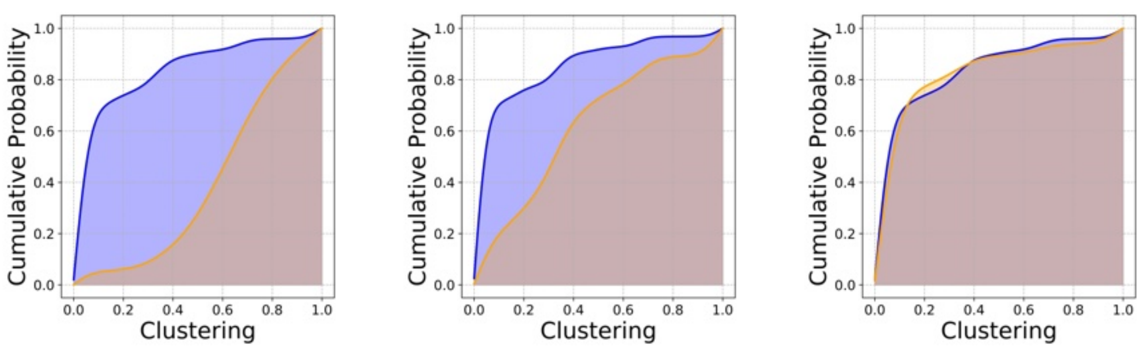
systems (Ravindran et al., 2023; Du et al., 2018). Excessive clustering traps energy, while under-clustering fragments stress pathways. Both impair mesoscale fidelity.

We compute the local clustering coefficient as:

$$C_i = \frac{2|\{(u,v) : u,v \in N(i), (u,v) \in \mathbb{E}\}|}{k_i(k_i - 1)}, \tag{39}$$

where k_i is the degree of node i and $N(i)$ its neighbor set.

Figure 21 shows that early-stage graphs are highly over-clustered: mean 0.5963 vs. 0.1014 in the real data, with median 0.6190 vs. 0.0. By epoch 250, generated graphs reproduce the target distribution: mean 0.1097 vs. 0.1014, with a median of 0.0 and standard deviation 0.2715 (vs. 0.2351). This alignment indicates that GraphRNN recovers realistic triadic heterogeneity without oversmoothing or collapse, enabling accurate modeling of localized stress networks.



(a) Epoch 10 (b) Epoch 50 (c) Epoch 250

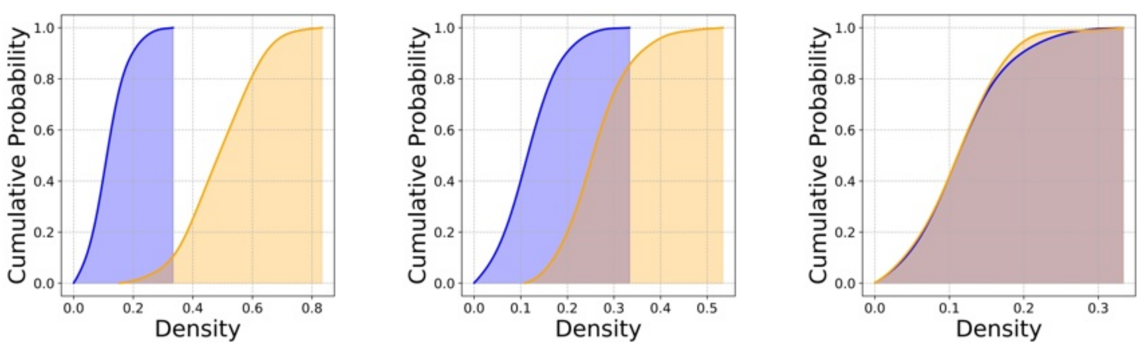
Fig. 21: Cumulative distribution of clustering coefficient over training. Initial over-aggregation is corrected over time.

5.2.4 *Global Connectivity Indicated by Graph Density*

Graph density summarizes global network compactness, which determines how fully grains connect across the PBX domain. In detonation modeling, density modulates energy dispersion and stress routing. Overdense graphs may exaggerate conduction pathways, while sparse ones risk underestimating load transfer (Roy et al., 2022; Ravindran et al., 2023; Du et al., 2018; Walley and Field, 2006). Graph density is defined as,

$$\text{Graph Density} = \frac{2N_E}{N_V(N_V-1)}, \tag{40}$$

where N_E and N_V are the number of edges and vertices. The graph density measures how connected the graph is. It reaches 1 when the graph becomes complete, i.e., all edges have been assigned. Figure 22 tracks the density distribution over training. At epoch 10, the graphs are overconnected, with a mean of 0.4865 compared to 0.1205 in the real data, and medians of 0.4844 and 0.1101, respectively. These values reflect pathologically dense structures that exaggerate cross-grain coupling. By epoch 250, density statistics converge: mean 0.1097 (vs. 0.1205), median 0.1105 (vs. 0.1101), and standard deviation 0.0596 (vs. 0.0631). This statistical data indicates that the synthetic and real microstructures are topologically quite close, although the generated microstructures in the sampling pool do not share the same probabilistic distribution as the benchmark.



(a) Epoch 10 (b) Epoch 50 (c) Epoch 250

Fig. 22: The eCDF of graph density across training. Initial overconnection is progressively corrected.

5.2.5 *Volumetric Phase Ratios and Material Composition*

The volume ratio between the energetic grain and polymer constituents is important for the performance of PBX (Guo et al., 2022), as it affects the pore collapse and the subsequent critical threshold velocity required for chemical reaction of detonation. As such, we compare the total grain volume and polymer-to-grain phase ratios between real and generated microstructures, assuming that the volume is occupied by either the polymeric binder or grains, thereby ignoring porosity for now. Given a fixed body with known volume, we can obtain the volume of the polymer binder by subtracting the volume of the grains. The polymer-to-grain ratio is therefore defined as

$$R_{pg} = \frac{V_{\text{polymer}}}{V_{\text{grain}}},$$

where V_{polymer} is the complement of total grain volume within the fixed RVE. Figure 23 presents the eCDF for both metrics. The synthetic graphs closely match the ground truth across the entire range, confirming that the model accurately replicates the global phase composition.

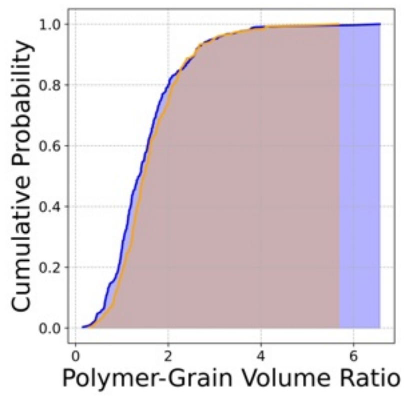


Fig. 23: eCDFs of polymer-to-grain volume fractions for real (blue) and generated (orange) microstructures.

The generated polymer-to-grain ratio has a mean of 1.5496 versus 1.5196 in the real data. The lower variance in the synthetic set indicates reduced extremal porosity, likely due to packing regularization during training. While this smoothing may marginally constrain structural heterogeneity, the full distributions remain well aligned. The GraphRNN–Diffusion pipeline preserves volumetric phase ratios with minimal deviation. Yet such morphological fidelity alone does not guarantee topological realism critical to mesoscale simulation.

6 Conclusions

Deriving, testing, and validating a constitutive model of high-fidelity requires a substantial amount of data. While these data often come from experiments, sub-scale simulations from representative elementary volumes become an attractive alternative when physical specimens are scarce and difficult to obtain. In this work, we leverage generative AI to create synthetic numerical specimens to address this problem by creating highly synthetic microstructures from mimicking the topologies and geometry of real microstructures obtained from micro-CT images. While there has already been a substantial amount of literature focusing on generating synthetic microstructures, overcoming the curse of high dimensionality remains a major challenge. Note that micro-CT images are often binarized for two-phase materials, such as PBX. As such, the convolution layer could be wasting a lot of FLOPs and bandwidth for the homogeneous space, where the embedding can only benefit from the highly concentrated gradients dominated at the interface of the constituents. In this work, we overcome the curse of high dimensionality by breaking the learning problems into two, i.e., the learning of topology and geometry. To achieve this goal, we incorporate GraphRNN, a generative AI designed to create realistic graphs that represent the granular network, and a conditional latent diffusion model synthesizes grain shapes conditioned on graph-level descriptors such as oriented bounding boxes and curvature. The results are validated using both topological and geometric measures across multiple length scales, ensuring that the synthetic microstructures generated by the two-scale generative AI are compatible with existing data. Future research directions may include fine-tuning frameworks that enhance the precision of enforcing conditions and controls on the generative process at both the topological and geometric levels through deep reinforcement learning, as well as controlling more complex performance measures, such as the effective elastic stiffness and damage criterion of the generated microstructures.

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8 CrediT authorship contribution statement

Jarett Poliner: Methodology, Software, Verification, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft. WaiChing Sun: Conceptualization, Methodology, Formal analysis, Resource, Writing – Original Draft, Supervision, Project administration, Funding acquisition. Khalid A. Alshibli: Data Curation, Resources, Formal analysis, Writing – Review & Editing; Richard Regueiro, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

9 Compliance with ethical standards

The authors declare that they have no conflict of interest.

References

Panos Achlioptas, Olga Diamanti, Ioannis Mitliagkas, and Leonidas Guibas. Learning representations and generative models for 3d point clouds. In *International conference on machine learning*, pages 40–49. PMLR, 2018.

Jhon Atchison and Sheng M. Shen. Logistic-normal distributions: Some properties and uses. *Biometrika*, 67 (2):261–272, 1980.

A. Barua, B. F. Henson, M. R. Armstrong, and M. Zhou. Ignition behavior of polymer-bonded explosives from microstructural stochasticity. *Journal of Applied Physics*, 113(18):184907, 2013. doi: 10.1063/1.4803543.

J. Beckvermit, J. Peterson, T. Harman, S. Bardenhagen, C. Wight, Q. Meng, and M. Berzins. Multiscale modeling of accidental explosions and detonations. *Computing in Science & Engineering*, 15(4):76–86, 2013.

S. Belon, B. Erzar, É. Kaeshammer, and L. Borne. Influence of the shape of rdx grains on the creation of hot spots by mesoscale modeling. *arXiv preprint arXiv:2012.00414*, 2020.

Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

Mario Botsch, Leif Kobbelt, Mark Pauly, Pierre Alliez, and Bruno L
'evy. *Polygon Mesh Processing*. CRC Press, 2010.

Chia-Tche Chang, Bastien Gorissen, and Samuel Melchior. Fast oriented bounding box optimization on the rotation group so(3, $\mathbb{R}$). *ACM Transactions on Graphics (TOG)*, 30(5):1–16, 2011.

Sehyun Chun, Sidhartha Roy, Yen Thi Nguyen, Joseph B Choi, Holavanahalli S Udaykumar, and Stephen S Baek. Deep learning for synthetic microstructure generation in a materials-by-design framework for heterogeneous energetic materials. *Scientific reports*, 10(1):13307, 2020.

Junyoung Chung, Caglar Gulcehre, KyungHyun Cho, and Yoshua Bengio. Empirical evaluation of gated recurrent neural networks on sequence modeling. *arXiv preprint arXiv:1412.3555*, 2014.

Andrew J Claydon. *Resonant acoustic mixing of polymer bonded explosives*. PhD thesis, 2021.

Akshay Dandekar and Marisol Koslowski. Effect of particle proximity and surface properties on the response of pbx under vibration. *Computational Materials Science*, 192:110334, 2021.

Taniya Das, Mark Koch, Maya Ravichandran, and Nikhil Khatri. Graphrnn revisited: An ablation study and extensions for directed acyclic graphs. *arXiv preprint arXiv:2307.14109*, 2023.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems*, 34:8780–8794, 2021.

James Diebel. Representing attitude: Euler angles, unit quaternions, and rotation vectors. *Matrix*, 58(15–16): 1–35, 2006.

Zongliang Du, Xiao-Yi Zhou, Renato Picelli, and H Alicia Kim. Connecting microstructures for multiscale topology optimization with connectivity index constraints. *Journal of Mechanical Design*, 140(11):111417, 2018.

Camilo A Duarte, Ahmed Hamed, Jonathan D Drake, Christian J Sorensen, Steven F Son, Weynong W Chen, and Marisol Koslowski. Void collapse in shocked-hmx single crystals: Simulations and experiments. *Propellants, Explosives, Pyrotechnics*, 45(2):243–253, 2020.

Mohammed Elnur and Khalid A Alshibli. Influence of x-ray beam exposure on the development of gas bubbles during triaxial testing of sand using 3d synchrotron micro-computed tomography. *Tomography of Materials and Structures*, 3:100016, 2023.

JW Feng, CF Li, S Cen, and DRJ Owen. Statistical reconstruction of two-phase random media. *Computers & Structures*, 137:78–92, 2014.

Silvia Ferrari and Francisco Cribari-Neto. Beta regression for modelling rates and proportions. *Journal of Applied Statistics*, 31(7):799–815, 2004.

J. E. Field. Hot spot ignition mechanisms for explosives. *Accounts of Chemical Research*, 25(11):489–496, 1992.

Stefan Aric Gottschalk. *Collision Queries Using Oriented Bounding Boxes*. PhD thesis, The University of North Carolina at Chapel Hill, Chapel Hill, NC, 2000.

Darla Graff Thompson, Bart Olinger, and Racci DeLuca. The effect of pressing parameters on the mechanical properties of plastic bonded explosives. *Propellants, Explosives, Pyrotechnics: An International Journal Dealing with Scientific and Technological Aspects of Energetic Materials*, 30(6):391–396, 2005.

Alex Graves. Generating sequences with recurrent neural networks. *arXiv preprint arXiv:1308.0850*, 2013.

Lori J Groven and Mark J Mezger. Printed energetics: The path toward additive manufacturing of munitions. In *Energetic Materials*, pages 115–128. CRC Press, 2017.

Kelly M Guan, Timothy I Anderson, Patrice Creux, and Anthony R Kovscek. Reconstructing porous media using generative flow networks. *Computers & Geosciences*, 156:104905, 2021.

Yuchen Guo, Rui Liu, Pengwan Chen, Bo Zhou, Gaoyang Hu, Chao Han, Kezhen Lv, and Shunpeng Zhu. Mechanical behavior of pbx with different hmx crystal size during die pressing: Experimental study and dem simulation. *Composites Science and Technology*, 222:109378, 2022.

Charles R Harris, K Jarrod Millman, Stéfan J van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J Smith, et al. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022.

Jonathan Ho et al. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 2020.

Jiangbei Hu, Ben Fei, Baixin Xu, Fei Hou, Weidong Yang, Shengfa Wang, Na Lei, Chen Qian, and Ying He. Topology-aware latent diffusion for 3d shape generation. *arXiv preprint arXiv:2401.17603*, 2024.

Norman L. Johnson, Samuel Kotz, and Narayanaswamy Balakrishnan. *Continuous Univariate Distributions, Volume 2*, volume 2. John Wiley & Sons, 1995.

Michael Kazhdan, Matt Bolitho, and Hugues Hoppe. Poisson surface reconstruction. *Proceedings of the fourth Eurographics symposium on Geometry processing*, pages 61–70, 2006.

Seokpum Kim, Yaochi Wei, Yasuyuki Horie, and Min Zhou. Prediction of shock initiation thresholds and ignition probability of polymer-bonded explosives using mesoscale simulations. *Journal of the Mechanics* *and Physics of Solids*, 114:97–116, 2018.

Rachel Kohler, Camilo Duarte Cordon, and Marisol Koslowski. Simulating dynamic failure of polymer-bonded explosives under periodic excitation. 2017.

Phaedon S Koutsourelakis and George Deodatis. Simulation of multidimensional binary random fields with application to modeling of two-phase random media. *Journal of engineering mechanics*, 132(6):619–631, 2006.

Ilya Loshchilov, Frank Hutter, et al. Fixing weight decay regularization in adam. *arXiv preprint arXiv:1711.05101*, 5(5):5, 2017.

Ran Ma, WaiChing Sun, and Catalin R Picu. Atomistic-model informed pressure-sensitive crystal plasticity for crystalline hmx. *International Journal of Solids and Structures*, 232:111170, 2021.

Ran Ma, WaiChing Sun, Catalin R Picu, and Tommy Sewell. Domain partitioning material point method for simulating shock in polycrystalline energetic materials. *Computer Methods in Applied Mechanics and Engineering*, 404:115815, 2023.

Douglas E Meisenheimer, Mark L Rivers, and Dorthe Wildenschild. Optimizing pink-beam fast x-ray microtomography for multiphase flow in 3d porous media. *Journal of Microscopy*, 277(2):100–106, 2020.

Hossein Mirzaee and Serveh Kamrava. Inverse design of microstructures using conditional continuous normalizing flows. *Acta Materialia*, 285:120704, 2025.

Kevin P. Murphy. *Machine Learning: A Probabilistic Perspective*. MIT Press, Cambridge, MA, 2012. ISBN 978-0-262-01802-9.

Phong CH Nguyen, Nikolaos N Vlassis, Bahador Bahmani, WaiChing Sun, HS Udaykumar, and Stephen S Baek. Synthesizing controlled microstructures of porous media using generative adversarial networks and reinforcement learning. *Scientific reports*, 12(1):9034, 2022.

Alex Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. *International Conference on Machine Learning*, 2021.

Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. In *ICML 2022*, 2022.

Open3D Development Team. Open3D documentation: `PointCloud.get_oriented_bounding_box`. https://www.open3d.org/html/python_api/open3d.geometry.PointCloud.html#open3d.geometry.PointCloud.get_oriented_bounding_box, 2025. [Online; accessed 20-June-2025].

Á. Orosz, V. Angelidakis, and K. Bagi. Surface orientation tensor to predict preferred contact orientation and characterise the form of individual particles. *Powder Technology*, 394:312–325, 2021.

Joseph O'Rourke. Finding minimal enclosing boxes. *International Journal of Computer & Information Sciences*, 14:183–199, 1985.

Adam Paszke. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, 2019.

Mariya Popova, Mykhailo Shvets, Junier Oliva, and Olexandr Isayev. Molecularrnn: Generating realistic molecular graphs with optimized properties. *arXiv preprint arXiv:1905.13372*, 2019.

Ian R. Porteous. *Geometric Differentiation: For the Intelligence of Curves and Surfaces*. Cambridge University Press, 2001.

Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 652–660, 2017.

Nikhila Ravi, Jeremy Reizenstein, David Novotny, Taylor Gordon, Wan-Yen Lo, Justin Johnson, and Georgia Gkioxari. Accelerating 3d deep learning with pytorch3d. *arXiv preprint arXiv:2007.08501*, 2020.

S Ravindran, V Gupta, and A Kidane. Mechanism of hotspot formation in energetic materials under mild impact loading. *Journal of Dynamic Behavior of Materials*, 9(4):375–383, 2023.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10684–10695, 2022.

Shobhan Roy, Belinda P Johnson, Xuan Zhou, Yen T Nguyen, Dana D Dlott, and HS Udaykumar. Hot spot ignition and growth from tandem micro-scale simulations and experiments on plastic-bonded explosives. *Journal of Applied Physics*, 131(20), 2022.

Szymon Rusinkiewicz. Estimating curvatures and their derivatives on triangle meshes. In *Proceedings of the 2nd International Symposium on 3D Data Processing, Visualization, and Transmission (3DPVT)*, pages 486–493. IEEE, September 2004. doi: 10.1109/TDPVT.2004.1335277.

Masao Satake. A discrete-mechanical approach to granular materials. *International journal of engineering science*, 30(10):1525–1533, 1992.

SciPy Community. Scipy documentation: `scipy.spatial.transform.Rotation.as_euler`. https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.transform.Rotation.as_euler.html, 2025. [Online; accessed 20-June-2025].

CB Skidmore, DS Phillips, BW Asay, DJ Idar, PM Howe, and DS Bolme. Microstructural effects in pbx 9501 damaged by shear impact. In *AIP Conference Proceedings*, volume 505, pages 659–662. American Institute of Physics, 2000.

Gregory G. Slabaugh. Computing euler angles from a rotation matrix. http://www.gregslabaugh.net/publications/euler.pdf, 2000. Retrieved on August 6, 2000.

Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *International conference on machine learning*, pages 2256–2265. pmlr, 2015.

Yang Song et al. Score-based generative modeling through stochastic differential equations. *International Conference on Learning Representations*, 2020.

Srihari Sundar and Veera Sundararaghavan. Database development and exploration of process–microstructure relationships using variational autoencoders. *Materials Today Communications*, 25:101201, 2020.

AC Thompson, J Llacer, L Campbell Finman, EB Hughes, JN Otis, S Wilson, and HD Zeman. Computed tomography using synchrotron radiation. *Nuclear Instruments and Methods in Physics Research*, 222(1-2): 319–323, 1984.

J-F Thovert, F Yousefian, P Spanne, CG Jacquin, and Pierre M Adler. Grain reconstruction of porous media: application to a low-porosity fontainebleau sandstone. *Physical Review E*, 63(6):061307, 2001.

A. C. Van der Steen, H. J. Verbeek, and J. J. Meulenbrugge. Influence of rdx crystal shape on the shock sensitivity of pbxs. In *Ninth Symposium (International) on Detonation*, volume 1, 1989.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, volume 30, 2017.

Nikolaos N Vlassis and WaiChing Sun. Denoising diffusion algorithm for inverse design of microstructures with fine-tuned nonlinear material properties. *Computer Methods in Applied Mechanics and Engineering*, 413:116126, 2023.

Nikolaos N Vlassis, Puhan Zhao, Ran Ma, Tommy Sewell, and WaiChing Sun. Molecular dynamics inferred transfer learning models for finite-strain hyperelasticity of monoclinic crystals: Sobolev training and validations against physical constraints. *International Journal for Numerical Methods in Engineering*, 123 (17):3922–3949, 2022.

Nikolaos N Vlassis, WaiChing Sun, Khalid A Alshibli, and Richard A Regueiro. Synthesizing realistic sand assemblies with denoising diffusion in latent space. *International Journal for Numerical and Analytical Methods in Geomechanics*, 48(16):3933–3956, 2024.

S. M. Walley and J. E. Field. Crystal sensitivities of energetic materials. *Journal of Materials Science*, 41(22): 6725–6748, 2006.

Kun Wang and WaiChing Sun. Anisotropy of a tensorial bishop's coefficient for wetted granular materials. *Journal of Engineering Mechanics*, 143(3):B4015004, 2017.

Li-Yi Wei. Parallel poisson disk sampling. *Acm Transactions On Graphics (tog)*, 27(3):1–9, 2008.

Francis Williams, Teseo Schneider, Claudio Silva, Denis Zorin, Joan Bruna, and Daniele Panozzo. Deep geometric prior for surface reconstruction. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10130–10139, 2019.

Mian Xiao, Jarett Poliner, and WaiChing Sun. Geometric learning for computational mechanics part iv: Efficient mesh-based plasticity from a domain-specific foundation model. *Computer Methods in Applied Mechanics and Engineering*, 446:118310, 2025.

Zijiang Yang, Xiaolin Li, L Catherine Brinson, Alok N Choudhary, Wei Chen, and Ankit Agrawal. Microstructural materials design via deep adversarial learning methodology. *Journal of Mechanical Design*, 140(11):111416, 2018.

John D Yeager, W Lee Perry, and Amanda L Duque. Predicting the effects of thermally-induced density gradients on the hydrodynamic behavior of pbx 9502. In *AIP Conference Proceedings*, volume 2272. AIP Publishing, 2020.

Jiaxuan You et al. Graphrnn: Generating realistic graphs with deep auto-regressive model. In *International Conference on Machine Learning*, 2018.

Hengshuang Zhao, Li Jiang, Jiaya Jia, Philip HS Torr, and Vladlen Koltun. Pointweb: Enhancing local neighborhood features for point cloud processing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5565–5573, 2019.

Qian-Yi Zhou, Jaesik Park, and Vladlen Koltun. Open3d: A modern library for 3d data processing. *arXiv preprint arXiv:1801.09847*, 2018.

Yanqiao Zhu, Yuanqi Du, Yinkai Wang, Yichen Xu, Jieyu Zhang, Qiang Liu, and Shu Wu. A survey on deep graph generation: Methods and applications. In *Proceedings of the First Learning on Graphs Conference (LoG 2022), PMLR 198, Virtual Event*, 2022.